

On the Anatomy of Trend

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Author

Christian Kjaer

ORCID: [0009-0004-8429-8998](https://orcid.org/0009-0004-8429-8998)

Affiliation and postal address

Head of Liquid Markets

ATP

Kongens Vænge 8

3400 Hillerød

Denmark

Corresponding author

Christian Kjaer

ATP, Kongens Vænge 8, 3400 Hillerød, Denmark

Email: cke@atp.dk

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On the Anatomy of Trend

Abstract

Trend-following strategies are often described as convex strategies that earn positive average returns and provide crisis protection. This paper develops a closed-form Gaussian framework for determining when trend-following generates conditional convexity, unconditional timing profits, and protection for a particular portfolio. The framework separates drift exposure from signal-return timing and maps those components into empirically testable conditional-payoff and downside-beta predictions. In fixed-horizon crashes, trend-following approaches a perfect hedge only when its signal tracks the realized drawdown. Under short-memory dynamics, a sufficiently severe sustained drawdown eventually generates a positive trend payoff. The multivariate extension shows that portfolio protection depends on the alignment between trend allocations and the portfolio's exposures. Across 81 return series over 25 years, conditional payoff convexity and downside betas are strongly significant, whereas the unconditional timing component is economically positive but statistically fragile. Performance across historical drawdowns varies with the signal horizon, and an external trend index provides only partial and noisy protection for a conventional 60/40 portfolio. The results therefore qualify common empirical interpretations of trend-following. Strong conditional convexity does not by itself establish a reliable unconditional return premium, and crisis protection depends on both signal dynamics and exposure alignment.

Keywords: time-series momentum, trend-following, crisis alpha, convexity, half-space identities, tail hedging

JEL classification: C58, G11, G12, G17.

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1. Introduction

A large literature documents cross-sectional momentum (Jegadeesh and Titman, 1993; Asness et al., 2013), time-series momentum (Moskowitz et al., 2012), and trend-following performance across long samples (Hurst et al., 2017), while a related literature emphasizes the convex payoff shape and crisis performance of managed-futures strategies (Fung and Hsieh, 2001; Dao et al., 2017; Hamill et al., 2016; Greyserman and Kaminski, 2014). Although these properties are well documented, treating them as manifestations of a common crisis-alpha mechanism lacks a unified theoretical foundation. Existing evidence does not by itself establish when a trend signal should generate a convex conditional payoff, an unconditional timing profit, or protection for a particular portfolio. A framework that separates these outcomes is therefore needed to interpret the empirical evidence and its implications for portfolio construction.

This paper develops such a framework under a stationary Gaussian benchmark. A trend strategy takes the sign of a linear signal and applies it to the subsequent return. Exact half-space identities decompose its expected return into drift exposure and signal-return timing. The same decomposition holds conditional on the benchmark's realized cumulative return. These results provide a theoretical basis for distinguishing conditional convexity from unconditional timing profits and for identifying the signal-drawdown alignment required for crash protection.

The analysis considers two severe-drawdown regimes with different empirical interpretations. In the first, the realized benchmark return tends to minus infinity over a fixed evaluation horizon. When the trend signal tracks this drawdown, the conditional trend payoff becomes asymptotically linear in drawdown severity, with its limiting left-tail slope measuring hedge quality. In the second regime, the drawdown grows linearly with the horizon. Under stationary Gaussian returns with short-memory dependence, a sufficiently severe sustained drawdown eventually produces a positive expected trend payoff. The effect of lookback length on crash protection depends on serial dependence, which determines how quickly the trend position moves into the protective direction during a drawdown. The multivariate results extend both regimes to a portfolio setting. They show that portfolio protection requires both signal responsiveness to the drawdown and alignment between trend allocations and the protected portfolio's asset exposures. Stand-alone sleeve convexity is not sufficient.

The empirical analysis uses this framework to study 81 return series spanning equities, government bonds, commodities, and credit and inflation markets over 25 years. The signals use lookbacks of 1, 3, 6, and 12 months. Returns are volatility scaled using full-sample standard deviations, so the exercise compares payoff shapes and timing estimates on a common ex-ante scale. Conditional regressions reveal a strong payoff smile, and piecewise downside betas are consistently negative. By contrast, the unconditional timing component is positive on average but statistically fragile. Circular block-bootstrap inference confirms that conditional convexity is much more robust than the unconditional timing estimate.

Drawdown episodes and an external benchmark provide two further tests. Across historical drawdowns, faster signals tend to react more effectively in shallower episodes, while slower signals can benefit in the deepest episodes. The limited number and heterogeneity of these events do not support a universal optimal lookback. Against a conventional 60/40 portfolio, the SG Trend Index has a negative monthly downside beta, but the protection is partial and noisy. Longer-horizon point estimates become more negative, although joint bootstrap inference does not establish a monotone horizon ordering. The portfolio alignment result offers a possible explanation for incomplete protection, but an aggregate index regression cannot identify the underlying sleeve-level channels.

The paper's contribution is therefore both analytical and empirical. Analytically, it derives exact conditional expectations and crash asymptotics that separate distinct notions of trend-following performance. Empirically, it uses those distinctions to discipline the interpretation of common findings. Strong conditional convexity does not imply a precisely estimated unconditional return premium. Positive performance in selected drawdowns does not imply a perfect hedge. A negative downside beta for a trend index does not establish that every underlying sleeve protects the portfolio. These qualifications complement the evidence on time-series momentum ([Moskowitz et al., 2012](#)), long-run trend performance ([Lempérière et al., 2014](#); [Hurst et al., 2017](#)), momentum crashes ([Daniel and Moskowitz, 2016](#)), and evidence from cross-sectional momentum that volatility scaling can reduce crash exposure ([Barroso and Santa-Clara, 2015](#)). They also build on analytical work on dynamic trading strategies, moving-average signals, and the momentum risk premium ([Bruder and Gaussel, 2011](#); [Grebekov and Serror, 2014](#); [Jusselin et al., 2017](#); [Roncalli, 2018](#)).

The Gaussian framework provides a tractable benchmark that yields ex-

act conditional expressions and isolates the role of serial dependence. The resulting crash guarantees are model-dependent, and heavy tails, stochastic volatility, and nonlinear dependence can alter how a cumulative drawdown is distributed through time. The theory therefore provides a transparent benchmark for organizing empirical evidence rather than a complete description of extreme-market dynamics.

The remainder of the paper proceeds as follows. Section 2 introduces the model and the drift-timing decomposition. Section 5 derives conditional payoffs. Section 6 studies fixed-horizon and sustained drawdowns and develops the portfolio alignment results. Section 8 presents the empirical evidence. Section 9 concludes. The appendices contain the proofs, implementation details, and robustness results.

2. Model

Let $(X_t)_{t \in \mathbb{Z}}$ denote one-period *log* returns of a benchmark. The analysis rests on the following assumption.

Assumption 1 (Stationary Gaussian benchmark). $(X_t)_{t \in \mathbb{Z}}$ is a stationary Gaussian process with mean $\mathbb{E}[X_t] = \mu$, variance $\text{Var}(X_t) = \sigma^2 > 0$, and autocovariance function $\gamma(h) := \text{Cov}(X_t, X_{t+h})$ independent of t . For the infinite-horizon results of Section 6.2 we assume in addition *short memory*, $\sum_{h \in \mathbb{Z}} |\gamma(h)| < \infty$, with strictly positive long-run variance $V := \sum_{h \in \mathbb{Z}} \gamma(h) > 0$.

In particular $X_t \sim \mathcal{N}(\mu, \sigma^2)$ for every t . All theoretical results below are stated under this Gaussian assumption.

Fix a lookback $k \geq 1$ and define the k -period signal and binary position

$$S_t := \sum_{j=1}^k X_{t-j}, \quad D_t := \text{sign}(S_t) \in \{-1, +1\}. \quad (1)$$

We study the per-period *log payoff*

$$Y_t := D_t X_t. \quad (2)$$

The signal is Gaussian $S_t \sim \mathcal{N}(k\mu, \sigma_S^2)$ with variance

$$\sigma_S^2 = \text{Var}(S_t) = \sum_{j=1}^k \sum_{\ell=1}^k \gamma(j - \ell). \quad (3)$$

We exclude degenerate signals, so $\sigma_S^2 > 0$. Whenever we condition on $B_T = b$, we also assume

$$\text{Var}(B_T) > 0, \quad \text{Var}(S_t | B_T) = \sigma_S^2 - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)} > 0,$$

for the dates under consideration.

The covariance c_{sx} follows directly from linearity of covariance and stationarity:

$$c_{sx} = \text{Cov}\left(\sum_{j=1}^k X_{t-j}, X_t\right) = \sum_{j=1}^k \text{Cov}(X_{t-j}, X_t) = \sum_{j=1}^k \gamma(j). \quad (4)$$

The unconditional average position of the strategy follows from $S_t \sim \mathcal{N}(k\mu, \sigma_S^2)$, since $\Pr(S_t > 0) = \Phi(k\mu/\sigma_S)$:

$$\mathbb{E}[D_t] = 2 \Pr(S_t > 0) - 1 = 2 \Phi\left(\frac{k\mu}{\sigma_S}\right) - 1. \quad (5)$$

For a finite horizon $T \geq 1$ we denote the benchmark cumulative log return by

$$B_T := \sum_{t=1}^T X_t. \quad (6)$$

Its variance and covariances with X_t and S_t can be expressed in terms of γ as

$$\text{Var}(B_T) = \sum_{u=1}^T \sum_{v=1}^T \gamma(u-v), \quad (7)$$

$$\text{Cov}(X_t, B_T) = \sum_{u=1}^T \gamma(t-u), \quad (8)$$

$$\text{Cov}(S_t, B_T) = \sum_{j=1}^k \text{Cov}(X_{t-j}, B_T) = \sum_{j=1}^k \sum_{u=1}^T \gamma(t-j-u). \quad (9)$$

Table 1 collects the notation used throughout.

Table 1: Notation	
Symbol	Meaning
X_t	one-period log excess return of the benchmark
μ, σ^2	mean and variance of X_t
$\gamma(h)$	autocovariance of (X_t) at lag h
k	lookback horizon of the trend signal
$S_t = \sum_{j=1}^k X_{t-j}$	trend signal, Gaussian variance σ_S^2
$D_t = \text{sign}(S_t) \in \{-1, +1\}$	binary trend position
$Y_t = D_t X_t$	per-period log payoff
$c_{sx} = \text{Cov}(S_t, X_t)$	signal-return covariance
T	evaluation / holding horizon
$B_T = \sum_{t=1}^T X_t$	cumulative benchmark log return
$m_x^{(t)}(b), m_s^{(t)}(b)$	conditional means of X_t, S_t given $B_T = b$
$v_x^{(t)}(b), v_s^{(t)}(b), c_{sx}^{(t)}(b)$	conditional (co)variances given $B_T = b$
$\mathbf{X}_t, P_t = w^\top \mathbf{X}_t$	asset-return vector and protected-portfolio return
$B_T^P = \sum_{t=1}^T P_t, Y_t^P$	cumulative protected-portfolio return and trend-programme payoff
$\beta_{i,t}^P, c_{sB,i,t}^P, \beta_{i,T}^P$	date-level return beta, signal-benchmark covariance, and finite-horizon portfolio beta
$\bar{\beta}_i^P$	long-run portfolio beta of sleeve i
Φ, ϕ	standard normal cdf and pdf

3. Gaussian half-space identities

The following identities are the analytic engine of the paper.

Lemma 1 (Gaussian half-space identities). *For any bivariate normal pair (X, S) with means (m_x, m_s) , variances (v_x, v_s) and covariance c_{sx} ,*

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right), \quad (10)$$

$$\mathbb{E}[X \text{sign}(S)] = 2 \mathbb{E}[X \mathbb{1}\{S > 0\}] - m_x. \quad (11)$$

These will be used repeatedly with $(X, S) = (X_t, S_t)$ and, in the conditional setting, with $(X, S) = (X_t, S_t)$ given $B_T = b$. A detailed derivation is provided in [Appendix A](#).

4. Expected Log Return

4.1. One-Period Results

Applying the Gaussian half-space identities (Lemma 1) with $(X, S) = (X_t, S_t)$ and the unconditional moments from Section 2 yields the following closed form.

Proposition 1 (Unconditional payoff and drift-timing decomposition). *Under Assumption 1, the expected per-period log payoff is*

$$\mathbb{E}[Y_t] = \mathbb{E}[D_t X_t] = \mathbb{E}[X_t \text{sign}(S_t)] = \mu \left(2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1 \right) + 2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right) \quad (12)$$

where σ_S^2 and c_{sx} are determined by γ as in Section 2. Equivalently, via the identity $\mathbb{E}[D_t X_t] = \mathbb{E}[D_t] \mathbb{E}[X_t] + \text{Cov}(D_t, X_t)$, the payoff splits into a directional component (drift μ times the average position $\mathbb{E}[D_t] = 2\Phi(k\mu/\sigma_S) - 1$) and a timing component $2 \frac{c_{sx}}{\sigma_S} \phi(k\mu/\sigma_S)$.

The proof is given in [Appendix A.1](#).

The first term in (12) is the unconditional drift of the underlying multiplied by the strategy's average position. It is therefore the strategy's directional exposure to the benchmark drift: when $\mu > 0$, the signal is on average long and the trend rule participates mechanically in the positive benchmark drift.

The second term is the timing component. It is proportional to c_{sx}/σ_S , the covariance between the lagged signal and the next-period return scaled by signal volatility, and is positive when the signal tends to be aligned with the subsequent return. The Gaussian weight $2\phi(k\mu/\sigma_S)$ is largest when the signal mean is close to the trading threshold and shrinks as the rule becomes strongly biased long or short.

4.2. Multi-Period Results

For horizon $T \geq 1$, define the cumulative log payoff

$$\sum_{t=1}^T Y_t = \sum_{t=1}^T D_t X_t. \quad (13)$$

Corollary 1 (Multi-period payoff). *Under Assumption 1, by stationarity of (D_t, X_t) ,*

$$\mathbb{E}\left[\sum_{t=1}^T Y_t\right] = T \mathbb{E}[D_t X_t] = T \left(\mu \left(2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1 \right) + 2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right) \right) \quad (14)$$

with c_{sx} and σ_S^2 determined by the autocovariance function γ .

When the underlying asset has a positive drift, the trend strategy will thus also yield a positive return even when there is no positive contribution from the timing component.

5. Conditional Multi-Period Expected Returns

To analyze how the performance of a trend strategy depends upon the underlying, we will fix a realized benchmark return $B_T = b$, and study the performance of the trend strategy given some b . Under the stationary Gaussian model, the vector (X_t, S_t, B_T) is jointly normal, so conditioning on $B_T = b$ preserves normality of (X_t, S_t) with computable conditional moments. The Gaussian half-space identities from Section 3 then apply with conditional means, variances and covariances. A detailed derivation of the conditional parameters and the extension of the half-space identities to the conditional setting is provided in [Appendix A.2](#).

5.1. Conditional parameters under a general Gaussian model

Using the autocovariance $\gamma(h)$, the conditional means and (co)variances of (X_t, S_t) given $B_T = b$ depend on t . We denote them by:

$$m_x^{(t)}(b) := \mathbb{E}[X_t \mid B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} (b - T\mu), \quad (15)$$

$$m_s^{(t)}(b) := \mathbb{E}[S_t \mid B_T = b] = k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} (b - T\mu), \quad (16)$$

$$v_x^{(t)}(b) := \text{Var}(X_t \mid B_T = b) = \sigma^2 - \frac{\text{Cov}(X_t, B_T)^2}{\text{Var}(B_T)}, \quad (17)$$

$$v_s^{(t)}(b) := \text{Var}(S_t \mid B_T = b) = \text{Var}(S_t) - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}, \quad (18)$$

$$c_{sx}^{(t)}(b) := \text{Cov}(S_t, X_t \mid B_T = b) = \text{Cov}(S_t, X_t) - \frac{\text{Cov}(S_t, B_T) \text{Cov}(X_t, B_T)}{\text{Var}(B_T)}. \quad (19)$$

Here all covariances involving X_t or S_t depend on t and are determined by $\gamma(h)$ via the formulas in Section 2. Note that while the conditional means depend on the realized value b , the conditional variances and covariance ($v_x^{(t)}(b), v_s^{(t)}(b), c_{sx}^{(t)}(b)$) are deterministic (constant in b) for Gaussian systems, though they vary with t .

5.2. Conditional expectations

Proposition 2 (Conditional expected payoff). *Under Assumption 1, for any horizon $T \geq 1$ and level b , the conditional multi-period log payoff is*

$$\mathbb{E} \left[\sum_{t=1}^T Y_t \mid B_T = b \right] = \sum_{t=1}^T \left\{ m_x^{(t)}(b) \left[2\Phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) - 1 \right] + 2 \frac{c_{sx}^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) \right\} \quad (20)$$

with the conditional moments $m_x^{(t)}(b)$, $m_s^{(t)}(b)$, $v_s^{(t)}(b)$, $c_{sx}^{(t)}(b)$ given above.

The result follows by applying the half-space identities to the conditional bivariate normal pair $(X_t, S_t \mid B_T = b)$ for each t . The justification that $(X_t, S_t \mid B_T = b)$ is bivariate normal with the stated parameters and that the Gaussian half-space identities extend to this conditional setting is provided in [Appendix A.2](#). Equation (20) is the exact analog of the unconditional payoff (14) with conditional moments in place of unconditional ones: a conditional drift times conditional average position, plus a conditional timing term.

6. Hedge properties

This section records the key asymptotic features of the conditional multi-period log payoff under the general stationary Gaussian model. We distinguish between two regimes: a fixed horizon T with an arbitrarily deep crash ($b \rightarrow -\infty$), and an infinite horizon $T \rightarrow \infty$ with a linear drawdown.

6.1. Fixed Horizon: Deep Crash ($b \rightarrow -\infty$)

Fix $T \geq 1$ and let $B_T = b$ tend to $-\infty$. From Proposition 2,

$$\mathbb{E} \left[\sum_{t=1}^T Y_t \mid B_T = b \right] = \sum_{t=1}^T \left\{ m_x^{(t)}(b) \left[2\Phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) - 1 \right] + 2 \frac{c_{sx}^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) \right\} \quad (21)$$

The asymptotic behavior of each term is governed by the standardized conditional signal mean

$$u_t(b) = \frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} = \frac{k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)}(b - T\mu)}{\sqrt{\sigma_S^2 - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}}}, \quad (22)$$

where

$$\text{Cov}(S_t, B_T) = \sum_{j=1}^k \sum_{u=1}^T \gamma(t - j - u), \quad \text{Var}(B_T) = \sum_{u=1}^T \sum_{v=1}^T \gamma(u - v).$$

For large $|b|$,

$$m_s^{(t)}(b) \sim \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} b, \quad m_x^{(t)}(b) \sim \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} b. \quad (23)$$

Thus the sign of $\text{Cov}(S_t, B_T)$ determines whether the conditional signal is asymptotically short, long, or unchanged. Define

$$\begin{aligned} \mathcal{T}_+ &:= \{t : \text{Cov}(S_t, B_T) > 0\}, \quad \mathcal{T}_- := \{t : \text{Cov}(S_t, B_T) < 0\}, \\ \mathcal{T}_0 &:= \{t : \text{Cov}(S_t, B_T) = 0\}. \end{aligned}$$

For $t \in \mathcal{T}_+$, $u_t(b) \rightarrow -\infty$ and the period contributes $-m_x^{(t)}(b)$ up to an exponentially small error. For $t \in \mathcal{T}_-$, $u_t(b) \rightarrow +\infty$ and the contribution is $+m_x^{(t)}(b)$. For $t \in \mathcal{T}_0$, the signal-dependent position and timing factors are constant in b , while the average-position contribution can still vary linearly through $m_x^{(t)}(b)$. This gives the following consolidated expansion.

Theorem 1 (Deep-crash asymptotics and the perfect hedge). *Under Assumption 1, fix $T \geq 1$ and partition $\{1, \dots, T\}$ into $\mathcal{T}_+, \mathcal{T}_-, \mathcal{T}_0$ by the sign of $\text{Cov}(S_t, B_T)$. Then, as $b \rightarrow -\infty$,*

$$\begin{aligned} \mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] &= -\sum_{t \in \mathcal{T}_+} m_x^{(t)}(b) + \sum_{t \in \mathcal{T}_-} m_x^{(t)}(b) \\ &\quad + \sum_{t \in \mathcal{T}_0} m_x^{(t)}(b) \left[2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1\right] \\ &\quad + \sum_{t \in \mathcal{T}_0} 2 \frac{\text{Cov}(X_t, S_t)}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right) + \mathcal{O}(e^{-cb^2}) \end{aligned} \quad (24)$$

for some constant $c > 0$. In particular:

(i) (Perfect hedge.) If $\text{Cov}(S_t, B_T) > 0$ for all t , then

$$\mathbb{E}\left[\sum_t Y_t \mid B_T = b\right] = -b + \mathcal{O}(e^{-cb^2}),$$

so the strategy is an asymptotically perfect hedge.

(ii) (i.i.d. benchmark.) If (X_t) is i.i.d., then for $T \geq 2$,

$$\mathbb{E}[\sum_t Y_t \mid B_T = b] = b\left(\frac{2}{T}\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) - 1\right) + \mathcal{O}(e^{-cb^2}),$$

which is positive for sufficiently negative b . For $T = 1$ the payoff is exactly $b(2\Phi(\frac{\mu}{\sigma}\sqrt{k}) - 1)$, a loss when $\mu > 0$ and zero when $\mu = 0$.

The $\mathcal{T}_0$ timing terms in (24) are bounded constants and do not affect the left-tail slope. The exponentially small error comes from the Mills-ratio bound in Appendix B, Lemma 3.

If $\text{Cov}(S_t, B_T) > 0$ for all t , then $\sum_t m_x^{(t)}(b) = \mathbb{E}[B_T \mid B_T = b] = b$, so

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = -\sum_{t=1}^T m_x^{(t)}(b) + \mathcal{O}(e^{-cb^2}) = -b + \mathcal{O}(e^{-cb^2}). \quad (25)$$

For an i.i.d. benchmark, $\mathcal{T}_0 = \{1\}$, $\mathcal{T}_+ = \{2, \dots, T\}$, and $m_x^{(t)}(b) = b/T$. Since $\sigma_S = \sqrt{k}\sigma$, (24) gives

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = b\left(\frac{2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)}{T} - 1\right) + \mathcal{O}(e^{-cb^2}), \quad T \geq 2. \quad (26)$$

For $T \geq 2$ the coefficient is strictly negative, hence the payoff is positive for all sufficiently negative b . For $T = 1$ the exact formula above applies. When $\mu > 0$, the strategy is typically long before the crash and loses on impact. The same term illustrates a lookback trade-off: in the i.i.d. case, increasing k raises the drift component of expected return but weakens the deep-crash hedge.

6.2. Infinite Horizon: Linear Drawdown ($T \rightarrow \infty$)

Now let $T \rightarrow \infty$ with $B_T = Td$ for fixed $d < 0$. Under short memory, $\text{Var}(B_T) \sim VT$, while $\text{Cov}(X_t, B_T)$ and $\text{Cov}(S_t, B_T)$ are bounded uniformly in t . Hence, away from the boundary of the horizon,

$$m_x^{(t)}(Td) \rightarrow d, \quad m_s^{(t)}(Td) \rightarrow kd,$$

and the conditional variances and covariance converge to their unconditional counterparts. In particular,

$$u_t(Td) = \frac{m_s^{(t)}(Td)}{\sqrt{v_s^{(t)}(Td)}} \rightarrow \frac{kd}{\sigma_S}.$$

Substituting these limits into (20) gives the per-period limit

$$g(d) = \underbrace{d \left[2\Phi\left(\frac{kd}{\sigma_S}\right) - 1 \right]}_{\text{Trend component}} + \underbrace{2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{kd}{\sigma_S}\right)}_{\text{Timing component}}. \quad (27)$$

The per-period summand is uniformly bounded and converges to $g(d)$ away from the two boundaries. Absolute summability ensures that boundary dates contribute only $o(T)$. Consequently, the average per-period payoff converges to $g(d)$:¹

$$\mathbb{E} \left[\sum_{t=1}^T Y_t \mid B_T = Td \right] = T \cdot g(d) + o(T). \quad (28)$$

The trend component is positive for every $d < 0$, because the conditional signal is short on average. The timing component has the sign of c_{sx} , but it is bounded and decays for sufficiently large $|d|$.

Theorem 2 (Linear-drawdown limit and threshold property). *Suppose Assumption 1 holds with short memory ($\sum_h |\gamma(h)| < \infty$, $V > 0$). For a linear drawdown $b = Td$ with fixed $d < 0$, as $T \rightarrow \infty$,*

$$\mathbb{E} \left[\sum_{t=1}^T Y_t \mid B_T = Td \right] = T g(d) + o(T), \quad g(d) = d \left[2\Phi\left(\frac{kd}{\sigma_S}\right) - 1 \right] + 2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{kd}{\sigma_S}\right). \quad (29)$$

Moreover, if $c_{sx} \geq 0$ then $g(d) > 0$ for every $d < 0$. If $c_{sx} < 0$ there exists a threshold $d^* < 0$ such that $g(d) > 0$ for all $d < d^*$. Hence a sufficiently severe sustained drawdown always renders trend-following profitable, even when the process is locally mean-reverting.

The threshold claim follows because the trend component grows linearly in $|d|$, while the timing component is bounded and decays like $\phi(kd/\sigma_S) \sim e^{-(kd/\sigma_S)^2/2}$. The interchange of limit and sum is justified in Appendix B, Lemmas 2 and 4.

¹The required uniform bound and averaging argument are given in Appendix B, Lemmas 2 and 4.

Lookback and Hedging: Competing Effects. The lookback k enters the hedge only through the standardized drawdown kd/σ_S , which sets how far the conditional signal is driven short and hence how close the directional term $d[2\Phi(kd/\sigma_S) - 1]$ comes to its saturated value: the larger $|kd/\sigma_S|$, the more reliably the position is short during the crash and the stronger the hedge. Increasing k has two competing effects on this quantity. It accumulates more of the cumulative loss in the signal, so the numerator $k|d|$ grows linearly in k . However, it also widens the signal dispersion σ_S in the denominator. For i.i.d. returns the two effects combine to $|kd/\sigma_S| = \sqrt{k}|d|/\sigma$, so the hedge does deepen with the lookback, but only at rate $\sqrt{k}$ rather than linearly. Serial correlation tilts this balance through σ_S : positive autocorrelation inflates the signal variance (overlapping returns reinforce one another), which slows the growth of kd/σ_S in k , whereas negative autocorrelation (local mean reversion) deflates σ_S and accelerates it. Consequently, a longer lookback buys comparatively more hedging in mean-reverting environments, where each added period sharpens the standardized signal, whereas highly persistent (strongly trending) processes reward a shorter, faster-reacting lookback.

6.3. Hedging in a Portfolio Context: the Alignment Principle

The preceding results use the same asset as both the traded market and the protected benchmark. A diversified trend programme instead trades many markets and is held against a portfolio whose composition need not match the traded universe. The single-asset logic still applies sleeve by sleeve, but conditioning on a portfolio drawdown requires additional notation to keep track of how each sleeve's return and signal respond to that drawdown.

6.3.1. Setup and notation

Let $\mathbf{X}_t = (X_{1,t}, \dots, X_{n,t})^\top$ be a jointly stationary Gaussian vector process with mean vector $\mu = (\mu_1, \dots, \mu_n)^\top$ and cross-autocovariances $\Gamma(h) = \text{Cov}(\mathbf{X}_t, \mathbf{X}_{t+h}) \in \mathbb{R}^{n \times n}$. A portfolio with weights $w \in \mathbb{R}^n$ defines the per-period linear return benchmark $P_t = w^\top \mathbf{X}_t$, with mean $\mu_P = w^\top \mu$, and cumulative return $B_T^P = \sum_{t=1}^T P_t$.² The trend programme runs one sleeve

²Because $X_{i,t}$ are log returns, $w^\top \mathbf{X}_t$ is a linear log-return benchmark. It is not, in general, the exact log return of a discretely rebalanced portfolio.

per asset, with a fixed lookback k and the same binary rule as before,

$$S_{i,t} = \sum_{j=1}^k X_{i,t-j}, \quad D_{i,t} = \text{sign}(S_{i,t}), \quad Y_{i,t} = D_{i,t} X_{i,t}. \quad (30)$$

The programme uses strictly positive sleeve allocations normalized to sum to one,

$$a_i > 0 \quad \text{for every } i, \quad \sum_{i=1}^n a_i = 1. \quad (31)$$

Its per-period payoff is

$$Y_t^P = \sum_{i=1}^n a_i Y_{i,t} = \sum_{i=1}^n a_i D_{i,t} X_{i,t}. \quad (32)$$

Write $B_{i,T} = \sum_{t=1}^T X_{i,t}$ and define the finite-horizon portfolio beta

$$\beta_{i,T}^P := \frac{\text{Cov}(B_{i,T}, B_T^P)}{\text{Var}(B_T^P)}, \quad \sum_{i=1}^n w_i \beta_{i,T}^P = 1. \quad (33)$$

For each sleeve and date, define

$$\beta_{i,t}^P := \frac{\text{Cov}(X_{i,t}, B_T^P)}{\text{Var}(B_T^P)}, \quad c_{sB,i,t}^P := \text{Cov}(S_{i,t}, B_T^P), \quad (34)$$

where $\sigma_{S_i}^2 = \text{Var}(S_{i,t})$. These objects have direct single-asset analogues. The date-level beta $\beta_{i,t}^P$ determines the conditional mean return on date t , while $c_{sB,i,t}^P$ determines whether the signal saturates short or long. Their roles must be recorded separately because portfolio beta only aggregates the return loadings,

$$\sum_{t=1}^T \beta_{i,t}^P = \beta_{i,T}^P, \quad (35)$$

and therefore does not reveal whether the signal is correctly aligned on every date. In the single-asset case the corresponding aggregate beta equals one, so this distinction is less visible. We assume $\text{Var}(B_T^P) > 0$ and $\text{Var}(S_{i,t} \mid B_T^P) > 0$ for every sleeve and date considered.

6.3.2. Fixed horizon: deep portfolio crash

Theorem 3 (Multi-asset deep-crash asymptotics). *Under the multivariate setup above, fix $T \geq 2$ and condition on $B_T^P = b \rightarrow -\infty$. Then*

$$\mathbb{E}\left[\sum_{t=1}^T Y_t^P \mid B_T^P = b\right] = L_T b + \mathcal{O}(1), \quad (36)$$

where the general fixed-horizon slope is

$$\boxed{\begin{aligned} L_T &= a^\top \ell_T, \\ \ell_{i,T} &:= - \sum_{t: c_{sB,i,t}^P > 0} \beta_{i,t}^P + \sum_{t: c_{sB,i,t}^P < 0} \beta_{i,t}^P \\ &\quad + \left[2\Phi\left(\frac{k\mu_i}{\sigma_{S_i}}\right) - 1 \right] \sum_{t: c_{sB,i,t}^P = 0} \beta_{i,t}^P. \end{aligned}} \quad (37)$$

Two cases parallel Theorem 1.

- (i) (Perfect hedge.) If $c_{sB,i,t}^P > 0$ for every i, t , then $L_T = -a^\top \beta_T^P$, where $\beta_T^P = (\beta_{1,T}^P, \dots, \beta_{n,T}^P)^\top$. In particular, if $a = w$, then $a^\top \beta_T^P = w^\top \beta_T^P = 1$ by (33), and

$$\mathbb{E}\left[\sum_{t=1}^T Y_t^P \mid B_T^P = b\right] = -b + \mathcal{O}(1).$$

More generally, the same conclusion holds whenever $a^\top \beta_T^P = 1$.

- (ii) (i.i.d. vector benchmark.) Suppose $\mathbf{X}_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \Sigma)$ over time, with arbitrary contemporaneous covariance Σ . Let

$$\beta_i^P := \frac{\text{Cov}(X_{i,t}, P_t)}{\text{Var}(P_t)}, \quad s_i := \text{sign}(\beta_i^P).$$

Then

$$L_T^{\text{iid}} = \sum_{i: \beta_i^P \neq 0} a_i |\beta_i^P| \left[\frac{2}{T} \Phi\left(s_i \frac{\sqrt{k} \mu_i}{\sigma_i}\right) - 1 \right]. \quad (38)$$

For $T \geq 2$, this coefficient is strictly negative if at least one $\beta_i^P \neq 0$. Hence the expected programme payoff is positive for all sufficiently negative b . For $n = 1$ and $a_1 = w_1 = 1$, this reduces to Theorem 1(ii).

The date-level quantities play distinct roles: $\beta_{i,t}^P$ measures the return exposure to the portfolio drawdown, while $c_{sB,i,t}^P$ determines the limiting trend position. In the i.i.d. case, the first signal is entirely pre-sample, so $c_{sB,i,1}^P = 0$, whereas $\text{sign}(c_{sB,i,t}^P) = \text{sign}(\beta_i^P)$ for $t \geq 2$. This boundary date produces the $2\Phi(s_i \sqrt{k} \mu_i / \sigma_i) / T$ term in (38). The full proof is in [Appendix B](#).

6.3.3. Linear portfolio drawdown

Assume now that $\sum_{h \in \mathbb{Z}} \|\Gamma(h)\| < \infty$ and define

$$V_P := \sum_{h \in \mathbb{Z}} \text{Cov}(P_0, P_h) > 0, \quad v_i := \sum_{h \in \mathbb{Z}} \text{Cov}(X_{i,0}, P_h), \quad \bar{\beta}_i^P := \frac{v_i}{V_P}.$$

Thus $\bar{\beta}_i^P$ is asset i 's long-run beta to the protected portfolio and is the long-horizon limit of $\beta_{i,T}^P$. Let

$$m_i(d) := \mu_i + \bar{\beta}_i^P (d - \mu_P), \quad c_{sx,i} := \text{Cov}(S_{i,t}, X_{i,t}), \quad (39)$$

and define

$$g_i^P(d) := m_i(d) \left[2\Phi\left(\frac{k m_i(d)}{\sigma_{S_i}}\right) - 1 \right] + 2 \frac{c_{sx,i}}{\sigma_{S_i}} \phi\left(\frac{k m_i(d)}{\sigma_{S_i}}\right), \quad (40)$$

$$G_P(d) := \sum_{i=1}^n a_i g_i^P(d).$$

Theorem 4 (Multi-asset linear-drawdown limit and threshold). *Under the assumptions above, let $B_T^P = Td$ with fixed $d < 0$. As $T \rightarrow \infty$,*

$$\mathbb{E}\left[\sum_{t=1}^T Y_t^P \mid B_T^P = Td\right] = T G_P(d) + o(T). \quad (41)$$

If at least one $\bar{\beta}_i^P \neq 0$, then there exists $d^ < 0$ such that $G_P(d) > 0$ for every $d < d^*$. Moreover,*

$$G_P(d) \sim -d \sum_{i=1}^n a_i |\bar{\beta}_i^P| \quad \text{as } d \rightarrow -\infty. \quad (42)$$

This is the direct portfolio counterpart of Theorem 2. A portfolio drawdown shifts sleeve i 's interior conditional return mean toward $m_i(d)$ and its signal mean toward $k m_i(d)$. For sufficiently negative d , every sleeve with nonzero $\bar{\beta}_i^P$ therefore takes the direction that profits from its own conditional move. Sleeves with $\bar{\beta}_i^P = 0$ contribute only bounded terms, while the aligned sleeves grow linearly in $|d|$. Under the standing allocation assumption, one nonzero long-run beta is therefore enough for the threshold result.

6.3.4. Sequential alignment and allocation

Corollary 2 (Sequential long-horizon alignment). *Under the short-memory assumptions above, taking the fixed-horizon deep-crash slope L_T and subsequently letting $T \rightarrow \infty$ gives*

$$L_T \longrightarrow -\sum_{i=1}^n a_i |\bar{\beta}_i^P|, \quad \beta_{i,T}^P \longrightarrow \bar{\beta}_i^P. \quad (43)$$

The three portfolio limits answer different questions. Equation (37) takes $b \rightarrow -\infty$ at fixed T and retains date-specific boundary effects. Equation (41) takes $T \rightarrow \infty$ with $b = Td$ and retains the finite drawdown rate d . Equation (43) first takes the fixed- T deep-crash slope and then sends $T \rightarrow \infty$. Only the last two share the same $d \rightarrow -\infty$ slope. They are not the same limit at a fixed finite d .

The alignment principle. Consider an investor who wants a trend overlay to protect portfolio P , while the overlay may trade a much broader universe. A sleeve is useful as a direct crash hedge only when a drawdown in P both moves that sleeve's underlying return and drives its trend signal into the profitable direction. At a fixed horizon, these two requirements are recorded separately by the date-specific return loading $\beta_{i,t}^P$ and signal covariance $c_{sB,i,t}^P$. Stand-alone trend profitability or convexity is therefore not enough to guarantee protection for this particular portfolio.

The sequential long-horizon result makes the distinction transparent: sleeve i contributes the crash slope $-a_i |\bar{\beta}_i^P|$. Thus a sleeve with $\bar{\beta}_i^P \approx 0$ is approximately orthogonal to the protected portfolio and provides little first-order directional protection, even if it is attractive on a stand-alone basis. Allocating overlay risk to many such sleeves can dilute the hedge. They may nevertheless be valuable if their covariance with the other trend sleeves reduces total overlay volatility. The relevant comparison is consequently not crash slope alone, but crash slope per unit of overlay risk.

This trade-off can be measured by letting $Z_{i,T}$ denote sleeve i 's cumulative trend payoff and Ω_T denote the covariance matrix of these payoffs:

$$\begin{aligned} Z_T &:= (Z_{1,T}, \dots, Z_{n,T})^\top, & Z_{i,T} &:= \sum_{t=1}^T Y_{i,t}, \\ \Omega_T &:= \text{Cov}(Z_T), & \text{HE}_T(a) &:= \frac{\sqrt{\text{Var}(B_T^P)} |a^\top \ell_T|}{\sqrt{a^\top \Omega_T a}}. \end{aligned}$$

The hedge-efficiency measure $\text{HE}_T(a)$ compares the magnitude of the overlay's conditional crash slope with its own volatility. The factor $\sqrt{\text{Var}(B_T^P)}$ expresses the slope for a one-standard-deviation move in the protected benchmark. Since $b < 0$ and the conditional payoff behaves as $(a^\top \ell_T)b$, protection requires $a^\top \ell_T < 0$. Temporarily relaxing the standing allocation restrictions, the direction that maximizes this hedge efficiency is

$$a_T^* \propto -\Omega_T^{-1} \ell_T, \quad (44)$$

provided Ω_T is positive definite and $\ell_T \neq 0$. Because hedge efficiency is scale invariant, a positive unconstrained direction can be normalized to sum to one. Its components need not all be positive, however, so it may violate the standing requirement $a_i > 0$. In that case, the optimal feasible allocation differs from (44). Under stationarity and absolutely summable autocovariances of the sleeve-payoff vector $Y_t := (Y_{1,t}, \dots, Y_{n,t})^\top$, suppose additionally that $\Gamma_Y := \sum_{h \in \mathbb{Z}} \text{Cov}(Y_0, Y_h)$ is positive definite. Then $T^{-1}\Omega_T \rightarrow \Gamma_Y$ and $\ell_T \rightarrow -|\bar{\beta}^P|$. The corresponding unconstrained hedge-efficiency-maximizing allocation direction is therefore $a^* \propto \Gamma_Y^{-1} |\bar{\beta}^P|$. This rule favors sleeves with strong direct loadings on the protected portfolio while accounting for overlap in sleeve risks through Γ_Y . It is an unconstrained benchmark rather than a guaranteed feasible allocation.

Why a 60/40 is imperfectly hedged by a diversified programme. A conventional 60/40 portfolio carries the large majority of its risk in equities, so its long-run portfolio-beta vector $\bar{\beta}^P$ is concentrated in equity-correlated markets and is small for markets orthogonal to it (e.g. many commodities). By (43) a broadly diversified trend programme hedges the 60/40 primarily through its equity-correlated sleeves, while its orthogonal sleeves have no direct long-horizon slope contribution and may consume risk budget. The hedge is strongest when the traded universe is aligned with the portfolio one wishes to protect. This prediction is directly testable on the investable trend indices: a broadly diversified programme such as the SG Trend index should display a real but diluted downside hedge against a conventional, equity-driven 60/40, an empirical question we take up in Section 8.

7. ARMA(p,q) Specialization

The preceding sections require only that (X_t) be a stationary Gaussian process with autocovariance $\gamma(h)$ and that γ be short-memory (e.g. absolutely

summable) for the linear-drawdown asymptotics. A broad and important class of processes satisfying these conditions is given by stationary Gaussian ARMA(p,q) models, which have been widely used in financial econometrics (Campbell et al., 1997). Evidence of positive serial correlation in weekly and monthly US equity returns over 1962–1985, particularly among smaller stocks (Lo and MacKinlay, 1988), provides empirical motivation for studying trend strategies under such processes.

Assume now that (X_t) follows a Gaussian ARMA(p,q) process

$$X_t - \mu = \sum_{j=1}^p \phi_j (X_{t-j} - \mu) + \sum_{\ell=0}^q \theta_\ell \varepsilon_{t-\ell}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (45)$$

with the usual stationarity and invertibility conditions on the AR and MA polynomials. Under these assumptions:

- (X_t) is stationary and Gaussian with mean μ and variance σ^2 ,
- the autocovariance function $\gamma(h) = \text{Cov}(X_t, X_{t+h})$ is determined uniquely by the ARMA parameters and is absolutely summable (short memory),
- hence all the general results derived above (one-period formulas, conditional multi-period expression, fixed- T left-tail behavior for $T \geq 2$, and the linear-drawdown limits) apply directly with this ARMA autocovariance $\gamma(h)$.

In particular, the ARMA(p,q) specialization is “admissible” in the sense that it satisfies all structural requirements of the Gaussian framework: joint normality of (X_t, S_t, B_T) , stationarity, and the short-memory condition needed for the asymptotic analysis in Section 6. Plugging the ARMA autocovariance function $\gamma(h)$ into the formulas in Sections 2–6 yields explicit expressions for $\text{Var}(S_t)$, $\text{Cov}(S_t, X_t)$, $\text{Var}(B_T)$, $\text{Cov}(X_t, B_T)$ and $\text{Cov}(S_t, B_T)$ in terms of the ARMA parameters, and hence explicit versions of all central results for the ARMA(p,q) case.

7.1. Example: Fixed-Horizon Deep Crash for a Gaussian MA(1)

As a compact illustration, consider a Gaussian MA(1) benchmark

$$X_t - \mu = \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (46)$$

with the usual invertibility condition $|\theta| < 1$. The autocovariance function is

$$\begin{aligned}\gamma(0) &= (1 + \theta^2)\sigma_\varepsilon^2 =: \gamma_0 > 0, \\ \gamma(1) &= \gamma(-1) = \theta\sigma_\varepsilon^2 =: \gamma_1, \\ \gamma(h) &= 0 \quad \text{for } |h| \geq 2.\end{aligned}$$

Then

$$\text{Var}(B_T) = T\gamma_0 + 2(T-1)\gamma_1.$$

For $T \geq 2$, the covariances between X_t and B_T take the simple form

$$\begin{aligned}\text{Cov}(X_1, B_T) &= \text{Cov}(X_T, B_T) = \gamma_0 + \gamma_1, \\ \text{Cov}(X_t, B_T) &= \gamma_0 + 2\gamma_1, \quad 2 \leq t \leq T-1.\end{aligned}$$

Consequently, the conditional means admit the representation

$$\begin{aligned}m_x^{(t)}(b) &:= \mathbb{E}[X_t \mid B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)}(b - T\mu) \\ &= \alpha_t b + \mathcal{O}(1),\end{aligned}$$

$$\sum_{t=1}^T \alpha_t = 1.$$

with

$$\alpha_1 = \alpha_T = \frac{\gamma_0 + \gamma_1}{T\gamma_0 + 2(T-1)\gamma_1}, \quad \alpha_t = \frac{\gamma_0 + 2\gamma_1}{T\gamma_0 + 2(T-1)\gamma_1}, \quad 2 \leq t \leq T-1.$$

The signal-benchmark covariance has a simple sign pattern. Since only the first lag contributes,

$$\text{Cov}(S_1, B_T) = \gamma_1. \quad (47)$$

For $t \geq 2$, the signal contains at least one return inside B_T . The overlapping contribution dominates any negative pre-sample contribution because $\gamma_0 + 2\gamma_1 = (1 + \theta)^2\sigma_\varepsilon^2 > 0$. Hence

$$\text{Cov}(S_t, B_T) > 0, \quad t = 2, \dots, T. \quad (48)$$

This single sign pattern gives three regimes:

- **Positive MA(1) correlation** ($\theta > 0$). Here $\text{Cov}(S_t, B_T) > 0$ for all t , so Theorem 1 gives the perfect-hedge limit

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = -b + \mathcal{O}(e^{-cb^2}), \quad b \rightarrow -\infty,$$

for any fixed $T \geq 1$ and $k \geq 1$.

- **i.i.d. benchmark** ($\theta = 0$). One has $\mathcal{T}_0 = \{1\}$ and $m_x^{(t)}(b) = b/T$, so

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = b \left[\frac{2}{T} \Phi\left(\frac{\mu\sqrt{k}}{\sigma}\right) - 1 \right] + \mathcal{O}(e^{-cb^2}),$$

$$b \rightarrow -\infty, \quad T \geq 2.$$

The payoff is positive in a deep crash for $T \geq 2$ and can be negative for $T = 1$ when $\mu > 0$.

- **Locally mean-reverting MA(1)** ($\theta < 0$). Now $\mathcal{T}_- = \{1\}$ and $\mathcal{T}_+ = \{2, \dots, T\}$, hence

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = -b + 2m_x^{(1)}(b) + \mathcal{O}(e^{-cb^2})$$

$$= (2\alpha_1 - 1)b + \mathcal{O}(e^{-cb^2}), \quad b \rightarrow -\infty.$$

with

$$2\alpha_1 - 1 = -\frac{(T-2)(\gamma_0 + 2\gamma_1)}{T\gamma_0 + 2(T-1)\gamma_1} = -\frac{(T-2)(1+\theta)^2\sigma_\varepsilon^2}{T(1+\theta^2)\sigma_\varepsilon^2 + 2(T-1)\theta\sigma_\varepsilon^2}.$$

This slope is negative for $T \geq 3$, so the payoff is positive in a deep crash. For $T = 2$, $\alpha_1 = \alpha_2 = 1/2$ and the leading payoff cancels to zero. For $T = 1$, the payoff is asymptotically negative when the strategy is biased long.

Thus the MA(1) example makes the general result explicit: positive local correlation gives a perfect crash hedge, while local mean reversion only delays the hedge until the crash spans enough periods.

7.2. Numerical Illustrations

Figures 1–3 illustrate these theoretical findings. Figure 1 displays the unconditional expected log payoff as a function of the MA parameter θ and lookback k . As expected, positive serial correlation ($\theta > 0$) yields high payoffs (momentum), while strong mean reversion ($\theta < 0$) can lead to losses unless the lookback k is tuned to capture longer-term trends. Figure 2 shows the conditional payoff profile, highlighting the convexity ("smile") characteristic of trend strategies in favorable regimes. Finally, Figure 3 confirms the asymptotic analysis: for deep crashes ($b \ll 0$), the simulated conditional payoff matches the panel-specific theoretical linear limit shown in each legend, illustrating the slope derived from Theorem 1.

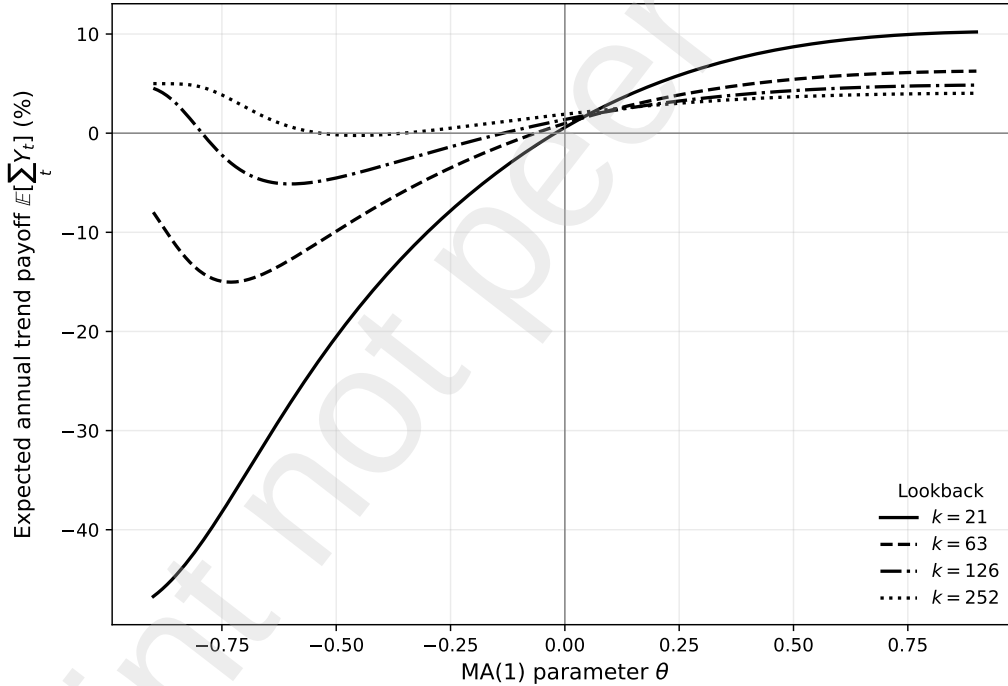


Figure 1: Unconditional expected annual trend payoff $\mathbb{E}[\sum_t Y_t]$ for a Gaussian MA(1) benchmark (annualized Sharpe ratio 0.5), as a function of the MA(1) parameter θ (horizontal axis), with one curve per lookback k . Positive serial correlation $\theta > 0$ (momentum) raises the payoff, whereas mean reversion $\theta < 0$ can produce losses unless k is lengthened. Curves are computed from the closed form (14).

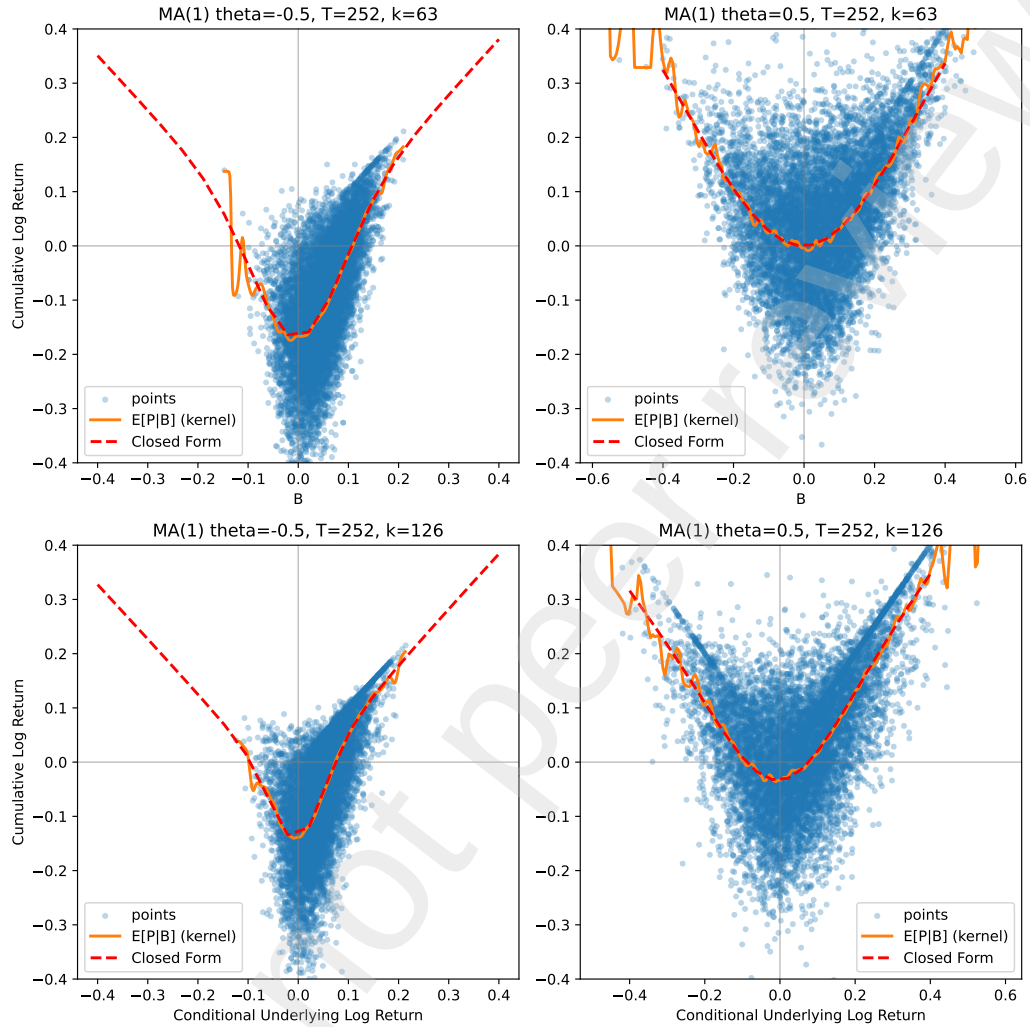


Figure 2: Conditional expected cumulative log payoff $\mathbb{E}[\sum_t Y_t \mid B_T = b]$ over $T = 252$ days versus the realized benchmark log return b (horizontal axis), for a Gaussian MA(1) benchmark (annualized Sharpe ratio 0.5) and several lookbacks k . The convex (“smile”) profile, low losses around $b \approx 0$ and gains in both tails, is the option-like signature of trend-following. Curves use the closed form (20).

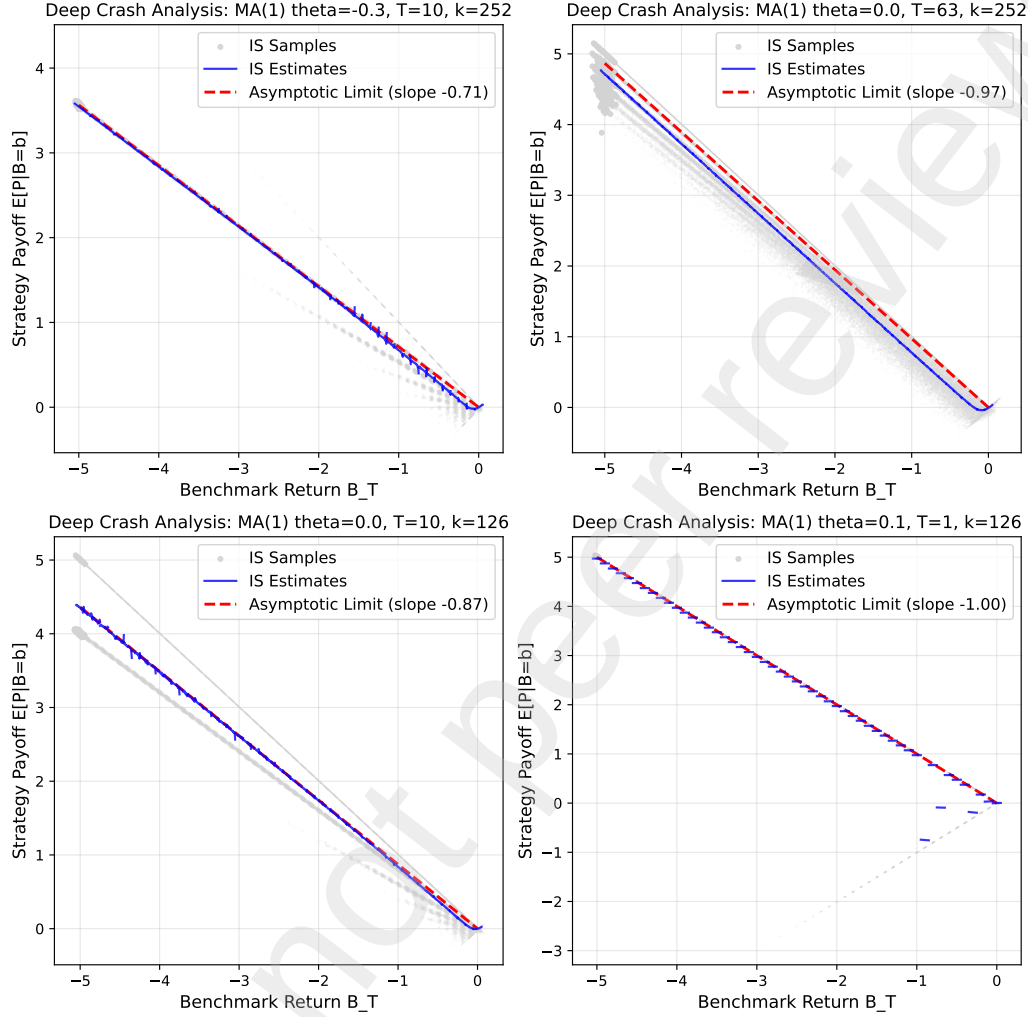


Figure 3: Fixed-horizon deep-crash limit for a Gaussian MA(1) benchmark. The conditional expected payoff $E[\sum_t Y_t | B_T = b]$ (solid) is plotted against the realized benchmark return b (horizontal axis). As $b \rightarrow -\infty$ it converges to the panel-specific theoretical linear asymptote (dashed, with slope shown in each legend), confirming the date-level sign partition in Theorem 1.

8. Empirical Analysis

This section tests the key theoretical predictions using a multi-asset dataset of daily returns. Our empirical analysis complements prior studies documenting trend-following performance across asset classes and time periods

(Moskowitz et al., 2012; Hurst et al., 2017; Lempérière et al., 2014).

8.1. Data and Implementation

The empirical sample contains 81 daily simple excess-return series for indices, futures, and par-bond or aggregate proxies across equity indices, government bonds, commodities, and credit/inflation markets. Each series is first calculated in local-currency excess-return terms. The daily excess return is then translated to DKK according to

$$R_{i,t}^{DKK} = R_{i,t}^{LC,ex} \frac{S_{i,t}}{S_{i,t-1}},$$

where $S_{i,t}$ denotes DKK per unit of the series' local currency. Expressing excess-return P&L in a common base currency provides a consistent basis for aggregating return contributions across markets when constructing multi-asset portfolios. The construction assumes a zero cross-currency basis. The sample runs from December 1999 to August 2025, with up to approximately 6,700 daily observations per series. For comparability, each series is normalized to 10% annualized volatility using its full-sample standard deviation (daily target $\sigma \approx 0.63\%$), after which the strategy calculations use $\log(1+R_t)$. This is a fixed full-sample normalization, not a real-time volatility forecast.

The trend strategy uses

$$S_t = \sum_{j=1}^k \log(1 + R_{t-j}), \quad D_t = \text{sign}(S_t).$$

Thus the position applied to day- t returns uses observations $t-k, \dots, t-1$, all available by the close of day $t-1$.

8.2. Statistical Significance of Timing

To test the contribution of the timing component, we utilize the theoretical decomposition derived in Eq. (12). By the identity $\mathbb{E}[DX] = \mathbb{E}[D] \mathbb{E}[X] + \text{Cov}(D, X)$, the total expected payoff decomposes as

$$\underbrace{\mathbb{E}[D_t X_t]}_{\text{Total Return}} = \underbrace{\mathbb{E}[D_t] \mathbb{E}[X_t]}_{\text{Static Drift Component}} + \underbrace{\text{Cov}(D_t, X_t)}_{\text{Timing Component}}, \quad (49)$$

where the first term is the return from simply holding the average position $\mathbb{E}[D_t]$, and the second term captures the additional return from aligning

the position with future returns. Under the Gaussian model of Section 2, this timing component equals the second term in Eq. (12), i.e. $2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right)$. Empirically, no distributional assumption is needed.

We test $H_0: \text{Cov}(D_t, X_t) = 0$ as follows. For each asset i and lookback k , define

$$\tau_{i,t}(k) := (D_{i,t}(k) - \bar{D}_i(k)) X_{i,t}, \quad \bar{D}_i(k) := \frac{1}{T} \sum_{t=1}^T D_{i,t}(k), \quad (50)$$

where $D_{i,t}(k) = \text{sign}(S_{i,t}(k))$ is the trend position based on lookback k . The sample mean $\frac{1}{T} \sum_t \tau_{i,t}(k)$ is the sample covariance $\widehat{\text{Cov}}(D_{i,t}(k), X_{i,t})$, so testing whether this mean is zero is a direct test of zero timing. Concretely, we regress

$$\tau_{i,t}(k) = \alpha_i(k) + \varepsilon_{i,t} \quad (51)$$

and test $H_0: \alpha_i(k) = 0$ using Newey–West HAC standard errors with k lags to account for the overlapping nature of the trend signal. The intercept $\alpha_i(k)$ is the timing component for asset i in daily return units.

For the aggregate, we use an intercept-only Fama–MacBeth procedure (Fama and MacBeth, 1973). At each date t , the cross-sectional regression on a constant has intercept

$$\bar{\tau}_t(k) := \frac{1}{N_t} \sum_{i=1}^{N_t} \tau_{i,t}(k), \quad (52)$$

where N_t is the number of histories with available data at date t . We average the daily intercepts and test $H_0: \mathbb{E}[\bar{\tau}_t(k)] = 0$ using Newey–West standard errors with k lags. This preserves contemporaneous cross-sectional dependence because all histories contribute to one coefficient at each date.

Table 2 reports the cross-sectional mean of the underlying return, the total trend return, and its decomposition into drift ($\bar{D}_i(k) \cdot \bar{X}_i$) and timing components. Timing is positive at every lookback, but none of the estimates is statistically significant at the 5% level: the largest Newey–West statistic is 1.9 at $k = 252$.

Table 2: Statistical Significance of Timing Component						
Lookback (k)	Annualized Averages				FM t-stat	% Sig. Indiv.
	Underlying	Trend	Drift	Timing		
21 Days	2.8%	1.9%	0.4%	1.5%	1.8	14%
63 Days	2.8%	1.6%	0.6%	1.0%	1.0	10%
126 Days	2.8%	1.6%	0.7%	0.9%	1.0	5%
252 Days	2.8%	2.5%	0.9%	1.6%	1.9	17%

Table 2 shows positive timing components at all lookbacks, but none has a two-sided Newey–West statistic above 1.96. The low individual significance rates underscore that timing is noisy at the single-history level, and the total trend return remains below the underlying despite the positive point estimates.

Inference robustness. The timing inference is qualitatively stable to Newey–West bandwidths of $k/2$ and $2k$. A circular block bootstrap of the Fama–MacBeth timing series (block length k , 5,000 resamples) gives explicitly one-sided p -values of 0.05, 0.16, 0.16 and 0.04 for $k = 21, 63, 126, 252$, respectively. These one-sided results provide some evidence of positive timing at one month and one year, but the conventional two-sided HAC evidence remains weak.

8.3. Transaction Costs and Turnover

The decomposition above is gross of trading frictions. Because the binary signal flips position, the strategy’s cost is governed by turnover, which falls sharply with the lookback. For each asset and lookback we measure turnover as the annualized number of round-trip position changes, $\frac{1}{2} |D_t - D_{t-1}| \times 252$, and we charge a one-way cost of c basis points of (volatility-scaled) notional per unit change in position, so that a full ± 1 reversal costs $2c$. Table 3 reports turnover and the net-of-cost annualized trend return at $c \in \{1, 2, 5\}$ bp, spanning the range of realistic costs for liquid futures.

Table 3: Turnover and Net-of-Cost Trend Return by Lookback					
Lookback (k)	Turnover (round trips/yr)	Annualized trend return			
		Gross	Net @1bp	Net @2bp	Net @5bp
21 Days	23.1	1.9%	1.5%	1.0%	-0.4%
63 Days	13.0	1.6%	1.4%	1.1%	0.3%
126 Days	8.7	1.6%	1.5%	1.3%	0.8%
252 Days	5.4	2.5%	2.4%	2.3%	2.0%

Turnover declines from roughly 23 round trips per year at $k = 21$ to about 5 at $k = 252$. Under the illustrative uniform cost assumptions of 1–2 bp, net returns remain positive across lookbacks. Only the fast signal turns marginally negative at 5 bp. Actual implementation costs differ across instruments.

8.4. Conditional Payoff Asymmetry

To test the non-linear payoff profile predicted by the theory, we estimate a piecewise linear regression for each lookback k and evaluation horizon T . At each date t , we run the cross-sectional regression

$$R_{i,t,T}^{\text{trend}}(k) = \alpha_t + \beta_{\text{down},t} \min(R_{i,t,T}^{\text{bench}}, 0) + \beta_{\text{up},t} \max(R_{i,t,T}^{\text{bench}}, 0) + \epsilon_{i,t}, \quad (53)$$

where $R_{i,t,T}^{\text{trend}}(k)$ and $R_{i,t,T}^{\text{bench}}$ are overlapping T -day strategy and benchmark returns. We report the Fama-MacBeth time-series means with Newey-West standard errors using $T + k$ lags. Portfolio-level regressions on an equal-weighted strategy, not reported, give the same qualitative patterns.

Table 4 reports the Fama-MacBeth mean coefficients $\beta_{\text{down}}/\beta_{\text{up}}$. The diagonal and near-diagonal cells ($T \approx k$) show the strongest asymmetry, with coefficients approaching unity, consistent with a straddle-like payoff at the strategy's natural horizon. The off-diagonal cells show the lookback trade-off: fast signals hedge long crashes imperfectly but react to short crashes, whereas long signals capture extended moves but are too slow for sudden shocks.

Table 4: Conditional Asymmetry Matrix ($\beta_{\text{down}}/\beta_{\text{up}}$)

Horizon (T)	Lookback (k)			
	21 Days	63 Days	126 Days	252 Days
21 Days	-0.61** / 0.82**	-0.18** / 0.53**	-0.01 / 0.45**	0.05 / 0.47**
63 Days	-0.59** / 0.73**	-0.66** / 0.94**	-0.22** / 0.69**	-0.07 / 0.62**
126 Days	-0.63** / 0.59**	-0.83** / 0.94**	-0.73** / 1.07**	-0.25 / 0.83**
252 Days	-0.50** / 0.50**	-0.76** / 0.78**	-0.93** / 1.01**	-0.62** / 1.05**

** indicates Fama-MacBeth t-statistic significant at the 1% level.

Subsample stability. To check whether the convexity is crisis-specific, we re-estimate the diagonal ($T = k$) smile coefficients on 1999–2012 and 2013–2025. Evaluation-return windows in the second subsample are required to lie fully within 2013–2025, while pre-2013 observations may initialize the signal. Table 5 shows the same sign pattern in both halves, although this descriptive

split is not an out-of-sample test because the fixed volatility normalization uses the full sample.

Table 5: Subsample Stability of the Diagonal ($T = k$) Smile ($\beta_{\text{down}}/\beta_{\text{up}}$)

Lookback ($k = T$)	Full sample	1999–2012	2013–2025
21 Days	-0.61 / 0.82	-0.64 / 0.86	-0.58 / 0.79
63 Days	-0.66 / 0.94	-0.71 / 0.98	-0.61 / 0.90
126 Days	-0.73 / 1.07	-0.67 / 0.99	-0.80 / 1.15
252 Days	-0.62 / 1.05	-0.29 / 0.99	-0.92 / 1.11

Table 6 reports the annualized Fama–MacBeth intercept α . The negative intercepts are consistent with an insurance-like cost of convexity. However, the intercept can also absorb drift, cross-asset heterogeneity, and departures from the piecewise-linear specification, so this interpretation is not separately identified.

Table 6: Annualized Piecewise-Regression Alpha

Horizon (T)	Lookback (k)			
	21 Days	63 Days	126 Days	252 Days
21 Days	-17.5%**	-9.0%**	-5.8%**	-3.8%**
63 Days	-9.3%**	-11.2%**	-6.9%**	-4.0%**
126 Days	-5.1%**	-8.3%**	-8.6%**	-4.6%**
252 Days	-2.7%**	-4.5%**	-6.2%**	-5.3%**

** indicates Fama-MacBeth t-statistic significant at the 1% level.

8.5. Drawdown Analysis

To describe performance in market stress, we identify peak-to-trough drawdowns of at least 5% in each full-sample-volatility-normalized excess-return history and compute trend returns from the start of each episode through its trough. These are drawdowns of the normalized excess-return indices, not unscaled asset-price drawdowns. Depth is measured as the ordinary percentage loss from the running peak. An event enters a lookback-specific average only when the complete strategy-return history is available. Table 7 is descriptive and reports the number of underlying episodes in each depth bucket.

Table 7: Average Trend Return during Drawdown Episodes (Depth vs. Lookback)

Drawdown Depth	Events	Lookback (k)			
		21 Days	63 Days	126 Days	252 Days
5-10%	545	-1.0%	-3.8%	-5.6%	-6.6%
10-20%	273	2.4%	-0.2%	-4.0%	-7.3%
20-40%	126	11.7%	9.3%	8.3%	5.6%
>40%	34	25.8%	27.3%	21.7%	32.1%

Events counts underlying episodes. Usable counts vary by lookback because complete signals are required.

The point estimates display a severity-dependent lookback trade-off: faster signals are less negative in shallow episodes, while the slowest signal has the largest mean in the deepest bucket. Because episodes are cross-sectionally and temporally dependent and no formal comparison is reported, these averages do not identify an optimal lookback or constitute a significance test of the theoretical threshold property.

8.6. Hedging a 60/40 Portfolio: A Word of Caution

The theory of Section 6.3 predicts that a diversified trend programme’s exact fixed-horizon hedge depends on date-specific alignment with the protected portfolio. In the sequential long-horizon deep-tail limit, sleeves with zero long-run portfolio covariance make no direct slope contribution. This mechanism suggests that a diversified programme may hedge an equity-driven 60/40 only imperfectly. We examine the aggregate relationship using the SG Trend Index (Bloomberg NEIXCTAT), regressing its cumulative h -month return on the matching cumulative return of a global 60/40 benchmark (60% global equities, 40% global bonds, built from the equity and rate aggregates of the dataset) in the same piecewise (down/up) form used above,

$$R_{t,h}^{\text{idx}} = \alpha + \beta_{\text{down}} \min(P_{t,h}^{60/40}, 0) + \beta_{\text{up}} \max(P_{t,h}^{60/40}, 0) + \epsilon_t. \quad (54)$$

The index is published monthly, so we evaluate horizons of 1, 3, 6, and 12 months. These are the monthly counterparts of the 21, 63, 126, and 252 trading-day grid in Table 4. We use overlapping h -month returns with Newey–West HAC standard errors of $\max(6, h + 2)$ lags. Because the SG Trend Index is a single aggregate programme rather than a cross-section of assets, we estimate one time-series regression rather than a Fama–MacBeth cross-sectional average. The SG Trend Index is a total-return series, so we

take it in excess of the 3-month US Treasury-bill rate to match the excess-return 60/40 benchmark. The intercept α is reported annualized. Both the excess index and the 60/40 are normalized separately to 10% annualized volatility using their full-sample standard deviations. Table 8 reports the coefficients $\beta_{\text{down}}/\beta_{\text{up}}$ and the alpha over 2000–2025.

Table 8: Crisis-hedge “smile” of the SG Trend Index against a global 60/40 ($\beta_{\text{down}}/\beta_{\text{up}}$ and annualized α), by evaluation horizon h . Piecewise regression $R_h^{\text{idx}} = \alpha + \beta_{\text{down}} \min(P_h^{60/40}, 0) + \beta_{\text{up}} \max(P_h^{60/40}, 0) + \epsilon$ on overlapping h -month returns, 2000–2025 ($n \approx 300$). The SG Trend total-return index is taken in excess of the 3-month US T-bill to match the excess-return 60/40, and both legs are normalized separately to 10% annualized volatility using their full-sample standard deviations. α is annualized.

Horizon (h)	$\beta_{\text{down}}/\beta_{\text{up}}$	α (ann.)
1 month	-0.29** / 0.10	-1.6%
3 months	-0.48** / 0.16	-1.4%
6 months	-0.51* / 0.09	0.2%
12 months	-0.61* / 0.04	1.4%

*, ** indicate Newey–West t -statistic significant at the 5%, 1% level.

The downside beta is negative and significant at every horizon, while the upside beta is small and insignificant. Its point estimate increases in magnitude from -0.29 at one month to -0.61 at twelve months. A joint 24-month moving-block bootstrap indicates that the three-month downside beta is more negative than the one-month beta at the 5% level. However, the 95% confidence intervals for the corresponding six-versus-one-month and twelve-versus-one-month differences include zero. The monotone point estimates are therefore suggestive of a deeper hedge over longer windows, not a statistically established ordering across all horizons. On this vol-normalized basis the downside beta remains well short of the one-for-one (-1) coefficient of the matched single-history diagonal in Table 4. The regression also contains an intercept and substantial residual risk, so -0.61 should not be read as deterministic offset.

The word of caution concerns this gap. At one month the hedge is partial ($\beta_{\text{down}} = -0.29$) and the month-to-month relationship is loose. Even the twelve-month point estimate is near -0.6 . The alignment principle offers a possible structural interpretation, because only sleeves with material portfolio covariance contribute directly to the long-run deep-tail slope. The aggregate regression does not observe the index’s sleeve weights or estimate sleeve-level portfolio betas, however, so it cannot attribute the gap to par-

ticular markets. The evidence supports describing the SG Trend Index as a genuine but partial and noisy hedge of this 60/40 benchmark.

9. Conclusion

This paper uses a stationary Gaussian benchmark to shed light on how empirical evidence on trend-following may be interpreted. The closed-form results distinguish conditional payoff convexity, unconditional signal-return timing, and protection of a particular portfolio. These properties are related, but none follows automatically from the others. The drift-timing decomposition identifies whether expected performance reflects average directional exposure or alignment between the signal and the subsequent return. Conditioning on the realized benchmark return extends this distinction to drawdowns and crisis hedging.

The empirical evidence is strongest for conditional payoff shape. Across 81 return series over 25 years, conditional payoff smiles and negative downside betas are statistically strong. By contrast, the unconditional timing component is positive on average but not statistically significant at conventional levels. Negative piecewise-regression intercepts are consistent with an insurance-like cost of convexity, although the regressions do not identify that interpretation causally. Historical drawdowns also show no universally superior lookback. Faster signals tend to respond more effectively to shorter episodes, while slower signals can benefit in the deepest and most persistent drawdowns.

The external benchmark produces a similarly qualified result. The SG Trend Index has a negative downside beta against a conventional 60/40 portfolio, but the protection is partial and noisy. Longer-horizon point estimates are more negative, although the joint bootstrap does not establish a monotone ordering across horizons. The portfolio theory provides one explanation for this result. Stand-alone convexity is insufficient for a complete hedge because protection also depends on the alignment between the trend programme's allocations and the portfolio's exposures.

The crash asymptotics make these qualifications explicit. Over a fixed horizon, the trend payoff approaches a perfect hedge only when the signal tracks the realized drawdown. Under short-memory Gaussian dynamics, a sufficiently severe sustained drawdown eventually produces a positive expected trend payoff. The multivariate results show that the same logic depends on portfolio alignment. These are conditional theoretical results rather

than distribution-free guarantees.

Heavy tails, stochastic volatility, and nonlinear dependence may change how extreme cumulative returns are distributed through time. The empirical event samples are also limited, and the aggregate trend-index regression cannot identify sleeve-level hedge contributions. Future work could examine the conditional predictions under richer return dynamics and test portfolio alignment using disaggregated trend allocations. The main empirical implication is that trend-following is more reliably characterized by conditional convexity and partial downside protection than by a precisely estimated unconditional timing premium.

Appendix A. Derivation of Gaussian Half-Space Identities

This appendix provides a detailed derivation of the Gaussian half-space identities stated in Section 3 and used throughout the paper. These identities allow us to compute expectations of the form $\mathbb{E}[X \mathbb{1}\{S > 0\}]$ and $\mathbb{E}[X \text{sign}(S)]$ when (X, S) is bivariate normal.

Appendix A.1. Unconditional Results

Appendix A.1.1. Setup

Let (X, S) be a bivariate normal random vector with means (m_x, m_s) , variances (v_x, v_s) , and covariance $c_{sx} = \text{Cov}(S, X)$. The correlation coefficient is

$$\rho = \frac{c_{sx}}{\sqrt{v_s v_x}}.$$

The joint density of (X, S) is

$$\begin{aligned} f_{X,S}(x, s) = & \frac{1}{2\pi\sqrt{v_x v_s}(1 - \rho^2)} \\ & \times \exp\left\{-\frac{1}{2(1 - \rho^2)} \left[\frac{(x - m_x)^2}{v_x} \right. \right. \\ & \left. \left. - 2\rho \frac{(x - m_x)(s - m_s)}{\sqrt{v_x v_s}} + \frac{(s - m_s)^2}{v_s} \right] \right\}. \end{aligned} \quad (\text{A.1})$$

Appendix A.1.2. Derivation of $\mathbb{E}[X \mathbb{1}\{S > 0\}]$

We want to compute

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = \int_{-\infty}^{\infty} \int_0^{\infty} x f_{X,S}(x, s) ds dx.$$

Step 1: Conditional expectation approach. We use the law of iterated expectations:

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = \mathbb{E}[\mathbb{E}[X | S] \mathbb{1}\{S > 0\}]. \quad (\text{A.2})$$

For bivariate normal (X, S) , the conditional distribution of X given $S = s$ is normal with mean and variance

$$\mathbb{E}[X | S = s] = m_x + \frac{c_{sx}}{v_s}(s - m_s) =: \mu_{X|S}(s), \quad (\text{A.3})$$

$$\text{Var}(X | S = s) = v_x - \frac{c_{sx}^2}{v_s} = v_x(1 - \rho^2) =: \sigma_{X|S}^2. \quad (\text{A.4})$$

Step 2: Integrate over $S > 0$. Using (A.2), we have

$$\begin{aligned} \mathbb{E}[X \mathbb{1}\{S > 0\}] &= \int_0^\infty \mathbb{E}[X | S = s] f_S(s) ds \\ &= \int_0^\infty \left(m_x + \frac{c_{sx}}{v_s}(s - m_s) \right) f_S(s) ds, \end{aligned} \quad (\text{A.5})$$

where $f_S(s) = \frac{1}{\sqrt{2\pi v_s}} \exp\left(-\frac{(s-m_s)^2}{2v_s}\right)$ is the marginal density of $S \sim \mathcal{N}(m_s, v_s)$.

Step 3: Split into two terms. We decompose (A.5) as

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \int_0^\infty f_S(s) ds + \frac{c_{sx}}{v_s} \int_0^\infty (s - m_s) f_S(s) ds.$$

The first integral is simply

$$\int_0^\infty f_S(s) ds = \Pr(S > 0) = \Phi\left(\frac{m_s}{\sqrt{v_s}}\right),$$

where Φ is the standard normal CDF.

Step 4: Evaluate the second integral. For the second integral, let $Z = (S - m_s)/\sqrt{v_s} \sim \mathcal{N}(0, 1)$. Then $s - m_s = \sqrt{v_s}Z$ and $ds = \sqrt{v_s}dZ$. The integral becomes

$$\begin{aligned} \int_0^\infty (s - m_s) f_S(s) ds &= \int_{-m_s/\sqrt{v_s}}^\infty \sqrt{v_s}Z \cdot \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ \\ &= \sqrt{v_s} \int_{-m_s/\sqrt{v_s}}^\infty Z \phi(Z) dZ, \end{aligned} \quad (\text{A.6})$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density.

Step 5: Standard Gaussian integral. Using the identity $Z\phi(Z) = -\frac{d}{dZ}\phi(Z)$, we have

$$\begin{aligned} \int_{-m_s/\sqrt{v_s}}^{\infty} Z\phi(Z) dZ &= - \int_{-m_s/\sqrt{v_s}}^{\infty} \frac{d\phi(Z)}{dZ} dZ \\ &= -[\phi(Z)]_{-m_s/\sqrt{v_s}}^{\infty} \\ &= -\left(0 - \phi\left(-\frac{m_s}{\sqrt{v_s}}\right)\right) \\ &= \phi\left(\frac{m_s}{\sqrt{v_s}}\right), \end{aligned} \quad (\text{A.7})$$

where we used $\phi(-z) = \phi(z)$ (symmetry of the standard normal density).

Step 6: Combine results. Substituting (A.7) into (A.6), we get

$$\int_0^{\infty} (s - m_s) f_S(s) ds = \sqrt{v_s} \phi\left(\frac{m_s}{\sqrt{v_s}}\right).$$

Therefore,

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right). \quad (\text{A.8})$$

Appendix A.1.3. Derivation of $\mathbb{E}[X \text{sign}(S)]$

Recall that $\text{sign}(S) = \mathbb{1}\{S > 0\} - \mathbb{1}\{S \leq 0\}$. By symmetry of expectations,

$$\begin{aligned} \mathbb{E}[X \text{sign}(S)] &= \mathbb{E}[X \mathbb{1}\{S > 0\}] - \mathbb{E}[X \mathbb{1}\{S \leq 0\}] \\ &= \mathbb{E}[X \mathbb{1}\{S > 0\}] - (\mathbb{E}[X] - \mathbb{E}[X \mathbb{1}\{S > 0\}]) \\ &= 2\mathbb{E}[X \mathbb{1}\{S > 0\}] - m_x. \end{aligned} \quad (\text{A.9})$$

Substituting (A.8) into (A.9), we obtain

$$\mathbb{E}[X \text{sign}(S)] = 2 \left(m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right) \right) - m_x. \quad (\text{A.10})$$

This can be rewritten more compactly as

$$\mathbb{E}[X \text{sign}(S)] = m_x \left(2\Phi\left(\frac{m_s}{\sqrt{v_s}}\right) - 1 \right) + 2\frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right), \quad (\text{A.11})$$

which is the form used in Section 4.1 of the main text.

Appendix A.1.4. Interpretation

The identity (A.11) decomposes the expected signed payoff into two components:

1. **Directional component:** $m_x(2\Phi(m_s/\sqrt{v_s}) - 1)$.

The term $2\Phi(m_s/\sqrt{v_s}) - 1 = \mathbb{E}[\text{sign}(S)]$ is the average position (probability of being long minus probability of being short). When multiplied by the unconditional mean m_x , this gives the expected return from simply holding the average position.

2. **Timing component:** $2\frac{c_{sx}}{\sqrt{v_s}}\phi(m_s/\sqrt{v_s})$.

This term captures the additional profit from timing: being more likely to be long when X is high and short when X is low. The covariance c_{sx} measures this alignment between the signal S and the return X . The Gaussian factor $\phi(m_s/\sqrt{v_s})$ weights this term, with maximum impact when $m_s \approx 0$ (signal mean near the threshold) and diminishing impact as $|m_s|$ grows large (strongly biased signal).

Appendix A.1.5. Alternative Derivation via Direct Integration

For completeness, we sketch an alternative direct derivation of (A.8). Starting from the joint density (A.1), we can complete the square in x to obtain

$$f_{X,S}(x, s) = f_S(s) \cdot f_{X|S}(x | s),$$

where $f_{X|S}(x | s)$ is the conditional density of X given $S = s$, which is $\mathcal{N}(\mu_{X|S}(s), \sigma_{X|S}^2)$. Then

$$\begin{aligned} \mathbb{E}[X \mathbb{1}\{S > 0\}] &= \int_0^\infty \int_{-\infty}^\infty x f_S(s) f_{X|S}(x | s) dx ds \\ &= \int_0^\infty f_S(s) \left(\int_{-\infty}^\infty x f_{X|S}(x | s) dx \right) ds \\ &= \int_0^\infty f_S(s) \mu_{X|S}(s) ds \\ &= \int_0^\infty f_S(s) \left(m_x + \frac{c_{sx}}{v_s}(s - m_s) \right) ds, \end{aligned} \quad (\text{A.12})$$

which recovers (A.5), and the rest of the derivation proceeds as before.

Appendix A.2. Conditional Results

The Gaussian half-space identities derived above extend naturally to conditional settings. In the main text (Section 5), we condition on the benchmark cumulative return $B_T = b$ and apply the identities to the conditional distribution of $(X_t, S_t \mid B_T = b)$. Here we establish the key properties that justify this approach.

Appendix A.2.1. Preservation of Normality

Consider the trivariate normal vector (X_t, S_t, B_T) under the stationary Gaussian model. Since $X_t, S_t = \sum_{j=1}^k X_{t-j}$, and $B_T = \sum_{u=1}^T X_u$ are all linear combinations of Gaussian returns $\{X_s\}$, the vector (X_t, S_t, B_T) is jointly normal with mean vector

$$\boldsymbol{\mu} = \begin{pmatrix} \mu \\ k\mu \\ T\mu \end{pmatrix}$$

and covariance matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) & \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 & \text{Cov}(S_t, B_T) \\ \text{Cov}(B_T, X_t) & \text{Cov}(B_T, S_t) & \text{Var}(B_T) \end{pmatrix}.$$

By the properties of the multivariate normal distribution, conditioning on $B_T = b$ yields a conditional bivariate normal distribution for $(X_t, S_t \mid B_T = b)$.

Appendix A.2.2. Conditional Parameters

The conditional mean vector and covariance matrix of (X_t, S_t) given $B_T = b$ are given by the standard formulas for conditional Gaussian distributions. Let

$$\mathbf{Z} = \begin{pmatrix} X_t \\ S_t \end{pmatrix}, \quad \boldsymbol{\mu}_1 = \begin{pmatrix} \mu \\ k\mu \end{pmatrix}, \quad \boldsymbol{\Sigma}_{11} = \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 \end{pmatrix},$$

and

$$\boldsymbol{\Sigma}_{12} = \begin{pmatrix} \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, B_T) \end{pmatrix}, \quad \Sigma_{22} = \text{Var}(B_T).$$

The conditional distribution is

$$(\mathbf{Z} \mid B_T = b) \sim \mathcal{N}(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}(b - T\mu), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}\boldsymbol{\Sigma}_{12}^\top).$$

Appendix A.2.3. Explicit Conditional Moments

Since $\Sigma_{22} = \text{Var}(B_T)$ is scalar, $\Sigma_{22}^{-1} = 1/\text{Var}(B_T)$. The conditional mean vector becomes

$$\mathbb{E}[X_t \mid B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)}(b - T\mu) =: m_x(b), \quad (\text{A.13})$$

$$\mathbb{E}[S_t \mid B_T = b] = k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)}(b - T\mu) =: m_s(b). \quad (\text{A.14})$$

The conditional covariance matrix is

$$\begin{aligned} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}^\top &= \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 \end{pmatrix} \\ &\quad - \frac{1}{\text{Var}(B_T)} \begin{pmatrix} \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, B_T) \end{pmatrix} \begin{pmatrix} \text{Cov}(X_t, B_T) & \text{Cov}(S_t, B_T) \end{pmatrix}. \end{aligned}$$

Computing each entry:

$$v_x(b) := \text{Var}(X_t \mid B_T = b) = \sigma^2 - \frac{\text{Cov}(X_t, B_T)^2}{\text{Var}(B_T)}, \quad (\text{A.15})$$

$$v_s(b) := \text{Var}(S_t \mid B_T = b) = \sigma_S^2 - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}, \quad (\text{A.16})$$

$$c_{sx}(b) := \text{Cov}(X_t, S_t \mid B_T = b) = \text{Cov}(X_t, S_t) - \frac{\text{Cov}(X_t, B_T) \text{Cov}(S_t, B_T)}{\text{Var}(B_T)}. \quad (\text{A.17})$$

Note that despite the notation $v_x(b)$, $v_s(b)$, $c_{sx}(b)$, these conditional variances and covariance are actually *constants* (independent of the realized value b) for Gaussian systems. Only the conditional means depend on b .

Appendix A.2.4. Application of Half-Space Identities

Once we have established that $(X_t, S_t \mid B_T = b)$ is bivariate normal with parameters $m_x(b)$, $m_s(b)$, $v_x(b)$, $v_s(b)$, $c_{sx}(b)$, the half-space identities derived earlier apply directly with these conditional parameters in place of

the unconditional ones. Specifically,

$$\mathbb{E}[X_t \mathbb{1}\{S_t > 0\} \mid B_T = b] = m_x(b) \Phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right) + \frac{c_{sx}(b)}{\sqrt{v_s(b)}} \phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right), \quad (\text{A.18})$$

$$\begin{aligned} \mathbb{E}[X_t \text{sign}(S_t) \mid B_T = b] &= 2\mathbb{E}[X_t \mathbb{1}\{S_t > 0\} \mid B_T = b] - m_x(b) \\ &= m_x(b) \left(2\Phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right) - 1 \right) + 2 \frac{c_{sx}(b)}{\sqrt{v_s(b)}} \phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right). \end{aligned} \quad (\text{A.19})$$

This is precisely the form used in equation (20) of Section 5.

Appendix A.2.5. Interpretation

The conditional half-space identities preserve the same two-component structure as the unconditional case:

1. **Conditional directional component:** $m_x(b)(2\Phi(m_s(b)/\sqrt{v_s(b)})-1)$ represents the conditional expected return $m_x(b)$ times the conditional average position $\mathbb{E}[\text{sign}(S_t) \mid B_T = b]$.
2. **Conditional timing component:** $2c_{sx}(b)/\sqrt{v_s(b)} \cdot \phi(m_s(b)/\sqrt{v_s(b)})$ captures the additional edge from the conditional alignment between signal and return.

The key insight is that conditioning on the cumulative benchmark return $B_T = b$ updates our beliefs about both the expected returns $m_x(b)$ and the signal strength $m_s(b)$, while the conditional volatilities remain well-defined constants. This allows the strategy's conditional performance to be analyzed using the same Gaussian half-space framework as the unconditional case.

Appendix B. Asymptotic Rigor

This appendix makes rigorous the interchange of limits and sums used in Section 6. Assume short memory, $\sum_{h \in \mathbb{Z}} |\gamma(h)| < \infty$, with long-run variance $V := \sum_h \gamma(h) > 0$.

Lemma 2 (Short-memory growth rates). *With short memory,*

$$\begin{aligned} \text{Var}(B_T) &= TV + o(T), \quad \sup_t |\text{Cov}(X_t, B_T)| \leq \sum_h |\gamma(h)| < \infty, \\ \sup_t |\text{Cov}(S_t, B_T)| &\leq k \sum_h |\gamma(h)| < \infty. \end{aligned}$$

Consequently $\beta_t = \text{Cov}(X_t, B_T) / \text{Var}(B_T) = \mathcal{O}(1/T)$ uniformly in t , and the conditional-variance reduction terms are $\mathcal{O}(1/T)$.

Proof. $T^{-1} \text{Var}(B_T) = \sum_{|h| < T} (1 - |h|/T) \gamma(h) \rightarrow V$ by dominated convergence, using absolute summability of γ . The covariance bounds follow from $\text{Cov}(X_t, B_T) = \sum_{u=1}^T \gamma(t - u)$ and the triangle inequality. $\square$

Lemma 3 (Deep-crash error bound). *Fix $T \geq 1$. For $b \rightarrow -\infty$, once the explicit $\mathcal{T}_0$ timing constants are retained as in (24), the remaining error in the left-tail expansion of Section 6.1 is $\mathcal{O}(e^{-cb^2})$ for some $c > 0$, uniformly over $t \in \mathcal{T}_+ \cup \mathcal{T}_-$.*

Proof. For $t \in \mathcal{T}_+ \cup \mathcal{T}_-$, $u_t(b) \rightarrow \mp\infty$ linearly in b . By the Mills-ratio bound $1 - \Phi(x) \leq \phi(x)/x$ for $x > 0$, both $|2\Phi(u_t(b)) - 1|$'s deviation from its limit ± 1 and $\phi(u_t(b))$ are exponentially small in b^2 . Multiplication by the affine conditional mean $m_x^{(t)}(b)$ preserves this bound after decreasing the exponent if necessary. Taking the minimum exponent over the finite index set gives the stated uniform remainder. For $t \in \mathcal{T}_0$, the position and timing factors are constant in b , but $m_x^{(t)}(b)$ need not be. The corresponding directional term is retained exactly in (24). $\square$

Lemma 4 (Linear-drawdown interchange). *Let $b = Td$ with $d < 0$. Under short memory, $\mathbb{E}[\sum_{t=1}^T Y_t \mid B_T = Td] = Tg(d) + o(T)$, where $g(d)$ is the per-period limit of Section 6.2.*

Proof. Write the per-period summand as

$$\begin{aligned} f_t(T) &= m_x^{(t)}(Td) [2\Phi(u_t(Td)) - 1] \\ &\quad + 2 \frac{c_{sx}^{(t)}(Td)}{\sqrt{v_s^{(t)}(Td)}} \phi(u_t(Td)). \end{aligned}$$

Lemma 2 gives $|\beta_t| \leq C/T$ for all sufficiently large T , so $|m_x^{(t)}(Td)| \leq |\mu| + C|d - \mu|$ uniformly in t . The conditional variance reduction is uniformly $\mathcal{O}(1/T)$, so the timing factor is uniformly bounded and hence $\sup_{t,T} |f_t(T)| < \infty$.

Let $r_t = \min(t - 1, T - t)$. Absolute summability implies that the differences between $\text{Cov}(X_t, B_T)$ and V , and between $\text{Cov}(S_t, B_T)$ and kV , are bounded by tails of $\sum_h |\gamma(h)|$ evaluated at r_t . Their Cesàro averages vanish because those tails tend to zero and only $o(T)$ dates have bounded r_t . Thus

$m_x^{(t)}(Td) \rightarrow d$, $u_t(Td) \rightarrow kd/\sigma_S$, and the conditional timing term converges to its unconditional counterpart in Cesàro mean. Continuity on the resulting uniformly bounded domain therefore gives

$$\frac{1}{T} \sum_{t=1}^T f_t(T) \rightarrow g(d),$$

which proves the claim. $\square$

Proof of Theorem 3. Condition on $B_T^P = b$. Since $(\mathbf{X}_t)_t$ is jointly Gaussian and $B_T^P = \sum_t w^\top \mathbf{X}_t$ is a linear functional, the triple $(X_{i,t}, S_{i,t}, B_T^P)$ is jointly Gaussian for every i, t . Conditioning on $B_T^P = b$ therefore places each sleeve i in the single-asset setting of Section 6.1, with the scalar benchmark B_T^P in the role of B_T . The relevant conditional means are

$$\begin{aligned} m_{x,i}^{(t)}(b) &= \mu_i + \beta_{i,t}^P(b - T\mu_P) = \beta_{i,t}^P b + \mathcal{O}(1), \\ m_{s,i}^{(t)}(b) &= k\mu_i + \frac{c_{sB,i,t}^P}{\text{Var}(B_T^P)}(b - T\mu_P). \end{aligned}$$

If $c_{sB,i,t}^P > 0$, the conditional position converges to -1 and the timing term is exponentially small. If $c_{sB,i,t}^P < 0$, it converges to $+1$. If $c_{sB,i,t}^P = 0$, the conditional signal distribution is unchanged, so the position mean equals $2\Phi(k\mu_i/\sigma_{S_i}) - 1$ and the timing term is bounded. Thus the coefficient of b in the conditional payoff of sleeve i is

$$- \sum_{t: c_{sB,i,t}^P > 0} \beta_{i,t}^P + \sum_{t: c_{sB,i,t}^P < 0} \beta_{i,t}^P + \left[2\Phi\left(\frac{k\mu_i}{\sigma_{S_i}}\right) - 1 \right] \sum_{t: c_{sB,i,t}^P = 0} \beta_{i,t}^P.$$

Multiplying by a_i and summing over sleeves proves (36) and (37).

For part (i), if $c_{sB,i,t}^P > 0$ for all i, t , then

$$\ell_{i,T} = - \sum_{t=1}^T \beta_{i,t}^P = -\beta_{i,T}^P$$

by (35). Hence $L_T = -a^\top \beta_T^P$, which becomes -1 when $a = w$ by (33), or more generally whenever $a^\top \beta_T^P = 1$.

For part (ii), let $\sigma_P^2 = \text{Var}(P_t)$ and $c_i = \text{Cov}(X_{i,t}, P_t)$. Temporal independence gives

$$\text{Var}(B_T^P) = T\sigma_P^2, \quad \beta_{i,t}^P = \frac{c_i}{T\sigma_P^2} = \frac{\beta_i^P}{T},$$

and

$$c_{sB,i,t}^P = \begin{cases} 0, & t = 1, \\ c_i \min(k, t - 1), & 2 \leq t \leq T. \end{cases}$$

Thus a sleeve with $\beta_i^P \neq 0$ has one zero-loading boundary date, followed by $T - 1$ dates on which $\text{sign}(c_{sB,i,t}^P) = s_i$. Hence

$$\begin{aligned} \ell_{i,T} &= \frac{|\beta_i^P|}{T} \left[s_i \left\{ 2\Phi\left(\frac{\sqrt{k}\mu_i}{\sigma_i}\right) - 1 \right\} - (T - 1) \right] \\ &= |\beta_i^P| \left[\frac{2}{T} \Phi\left(s_i \frac{\sqrt{k}\mu_i}{\sigma_i}\right) - 1 \right]. \end{aligned}$$

When $\beta_i^P = 0$, both $\beta_{i,t}^P$ and the slope contribution vanish. Summing over sleeves proves (38). For $T \geq 2$ and finite $\sqrt{k}\mu_i/\sigma_i$, the bracketed coefficient is strictly negative, which proves the positivity statement for a deep crash. $\square$

Proof of Theorem 4. Let $\gamma_{iP}(h) = \text{Cov}(X_{i,0}, P_h)$. Absolute summability gives

$$\text{Var}(B_T^P) = TV_P + o(T).$$

For dates whose distance from both boundaries tends to infinity,

$$\text{Cov}(X_{i,t}, B_T^P) \longrightarrow v_i, \quad \text{Cov}(S_{i,t}, B_T^P) \longrightarrow kv_i.$$

The corresponding average absolute deviations vanish by the same tail argument used in Lemma 4. Conditioning on $B_T^P = Td$ therefore yields, away from the boundaries,

$$\begin{aligned} \mathbb{E}[X_{i,t} \mid B_T^P = Td] &\longrightarrow \mu_i + \bar{\beta}_i^P(d - \mu_P) = m_i(d), \\ \mathbb{E}[S_{i,t} \mid B_T^P = Td] &\longrightarrow k m_i(d). \end{aligned}$$

The covariances with B_T^P are uniformly bounded. Their contributions to each conditional variance and covariance are consequently $\mathcal{O}(1/T)$, so

$$\text{Var}(S_{i,t} \mid B_T^P = Td) \longrightarrow \sigma_{S_i}^2, \quad \text{Cov}(S_{i,t}, X_{i,t} \mid B_T^P = Td) \longrightarrow c_{sx,i}.$$

The Gaussian half-space identity now gives the interior per-period limit $g_i^P(d)$ in (40). The summands are uniformly bounded for fixed d , and boundary

dates contribute only $o(T)$. Averaging over dates, summing over the finite set of sleeves, and multiplying by T proves (41).

If $\bar{\beta}_i^P \neq 0$, then as $d \rightarrow -\infty$,

$$m_i(d) \sim \bar{\beta}_i^P d, \quad m_i(d) \left[2\Phi\left(\frac{k m_i(d)}{\sigma_{S_i}}\right) - 1 \right] \sim |\bar{\beta}_i^P|(-d),$$

while the timing term converges to zero. If $\bar{\beta}_i^P = 0$, the entire sleeve contribution is constant in d . With at least one nonzero $\bar{\beta}_i^P$, the standing allocation assumptions imply that summing these limits proves (42) and the existence of d^* . $\square$

Proof of Corollary 2. Absolute summability gives

$$\text{Var}(B_T^P) = TV_P + o(T), \quad \sum_{t=1}^T \beta_{i,t}^P = \frac{\text{Cov}(B_{i,T}, B_T^P)}{\text{Var}(B_T^P)} = \beta_{i,T}^P \longrightarrow \bar{\beta}_i^P.$$

For a date whose distance from both endpoints diverges, $c_{sB,i,t}^P \rightarrow kv_i$, the average absolute deviation from this limit vanishes by the same tail argument as in Lemma 4. Moreover $\beta_{i,t}^P = \mathcal{O}(1/T)$ uniformly.

If $v_i \neq 0$, the set of dates on which $\text{sign}(c_{sB,i,t}^P) \neq \text{sign}(v_i)$, including those with $c_{sB,i,t}^P = 0$, has vanishing proportion. Hence the sleeve's summed slope in (37) converges to $-\text{sign}(v_i)\bar{\beta}_i^P = -|\bar{\beta}_i^P|$.

If $v_i = 0$, let $c_{i,t,T} := \text{Cov}(X_{i,t}, B_T^P)$. Absolute summability and the same averaging argument give $T^{-1} \sum_{t=1}^T |c_{i,t,T}| \rightarrow 0$. Since the absolute value of the average-position factor is at most one, the absolute value of the sleeve's slope is bounded by

$$\sum_{t=1}^T |\beta_{i,t}^P| = \frac{\sum_{t=1}^T |c_{i,t,T}|}{\text{Var}(B_T^P)} \longrightarrow 0.$$

Because $\bar{\beta}_i^P = 0$ in this case, (43) follows. $\square$

Remark 1 (Numerical verification of the closed-form results). The Gaussian half-space identities, single-asset and portfolio fixed-horizon slopes, zero-slope edge cases, linear-drawdown limits, short-memory growth rate, and distinction between direct and sequential portfolio limits were checked by symbolic computation, exact conditional calculations, and Monte Carlo simulation. The checks and their explicit tolerances are included in the replication code. They do not replace the analytical proofs, but provide independent protection against algebraic and implementation error.

Appendix C. Data Universe

Table C.1 lists the 81 return series included in the empirical study. The accompanying manifest records the full vendor names, coverage, and missingness.

Table C.1: Asset Universe (81 Return Series)

Asset Class	Instruments
Equities (34)	MSCI Australia, SPI 200 Fut, Hang Seng Fut, MSCI Hong Kong, MSCI Japan, Topix Fut, MSCI Singapore, SGX MSCI Singapore Fut, Ibovespa Fut, MSCI India, SGX Nifty 50 Fut, KOSPI 200 Fut, MSCI Mexico, MSCI Mexico Fut, MSCI South Africa, South Africa Fut, KAX, MSCI Denmark, IBEX 35 Fut, MSCI Spain, CAC 50 Fut, MSCI France, DAX 50 Fut, MSCI Germany, FTSE-MIB 40 Fut, MSCI Italy, MSCI Swiss, SMI 20 Fut, FTSE 100 Fut, MSCI UK, MSCI Canada, SPTSX 60 Fut, S&P 500, S&P 500 Fut
Government Bonds (34)	AUS 10Y Bond Fut, AUS 10Y Par Bond, AUS 3Y Bond Fut, AUS 3Y Par Bond, AUS 5Y Par Bond, JPN 10Y Bond Fut, JPN 10Y Par Bond, JPN 3Y Par Bond, JPN 5Y Par Bond, FR 10Y, 3Y and 5Y Par Bonds, GER 10Y Bond Fut, GER 10Y Par Bond, GER 2Y Bond Fut, GER 3Y Par Bond, GER 5Y Bond Fut, GER 5Y Par Bond, UK 10Y Bond Fut, UK 10Y, 3Y and 5Y Par Bonds, CAN 10Y Bond Fut, CAN 10Y Par Bond, CAN 3Y Par Bond, CAN Rates 5Y, CAN 5Y Bond Fut, CAN 5Y Par Bond, US 10Y Bond Fut, US 10Y Par Bond, US 2Y Bond Fut, US 3Y Par Bond, US 5Y Bond Fut, US 5Y Par Bond
Commodities (8)	Gold Fut, Aluminum Fut, Copper Fut, Nickel Fut, Zinc Fut, Oil, Brent Oil Fut, Crude Oil Fut
Credit/Inflation (5)	Euro High Yield, US High Yield, Euro/GER 7–10Y breakeven inflation, UK 7–10Y breakeven inflation, US 7–10Y breakeven inflation

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Data availability

The empirical analysis uses a proprietary multi-asset panel containing daily local-currency excess returns translated to DKK as described in the data section, together with the external SG Trend Index obtained through Bloomberg and the 3-month US Treasury-bill rate from FRED (series TB3MS).

The proprietary panel and Bloomberg data are subject to licensing restrictions and cannot be redistributed. Exact reproduction of the empirical results requires access to these licensed data. The FRED series is publicly available.

Code availability

Replication code and documentation are available to the editor and reviewers upon request and will be deposited in a public repository upon acceptance. The package contains the canonical empirical notebook, shared universe loader and manifest, pinned environment specification, Gaussian verification suite, theoretical figure generator, and robustness and benchmark-analysis scripts. Proprietary multi-asset and Bloomberg data cannot be redistributed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

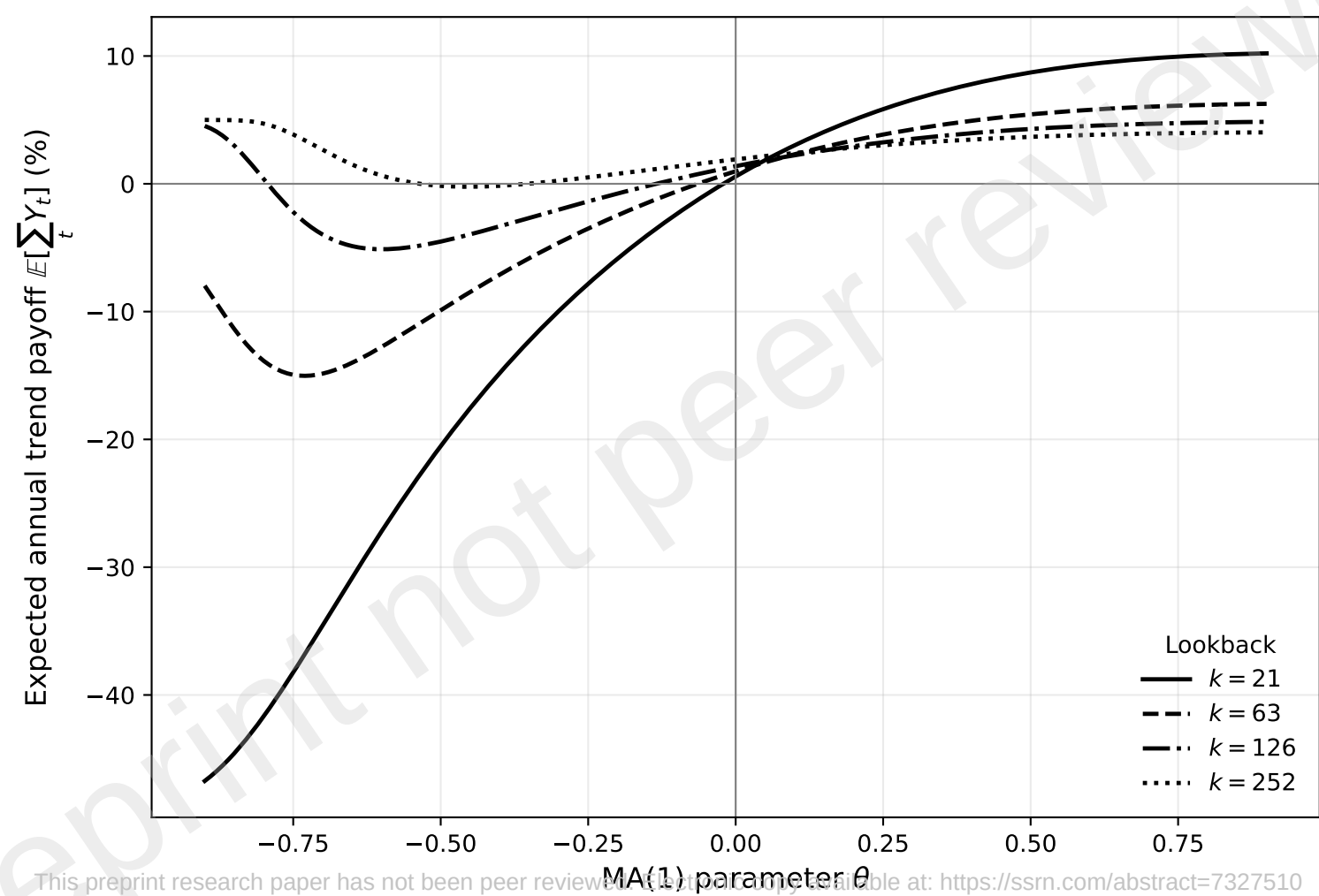
During the preparation of this work, the author used Anthropic Claude and OpenAI GPT models for editorial assistance, mathematical derivation support, reference checking, and code support. The author independently reviewed and verified all AI-assisted output, edited the content as needed, and takes full responsibility for the content of the article.

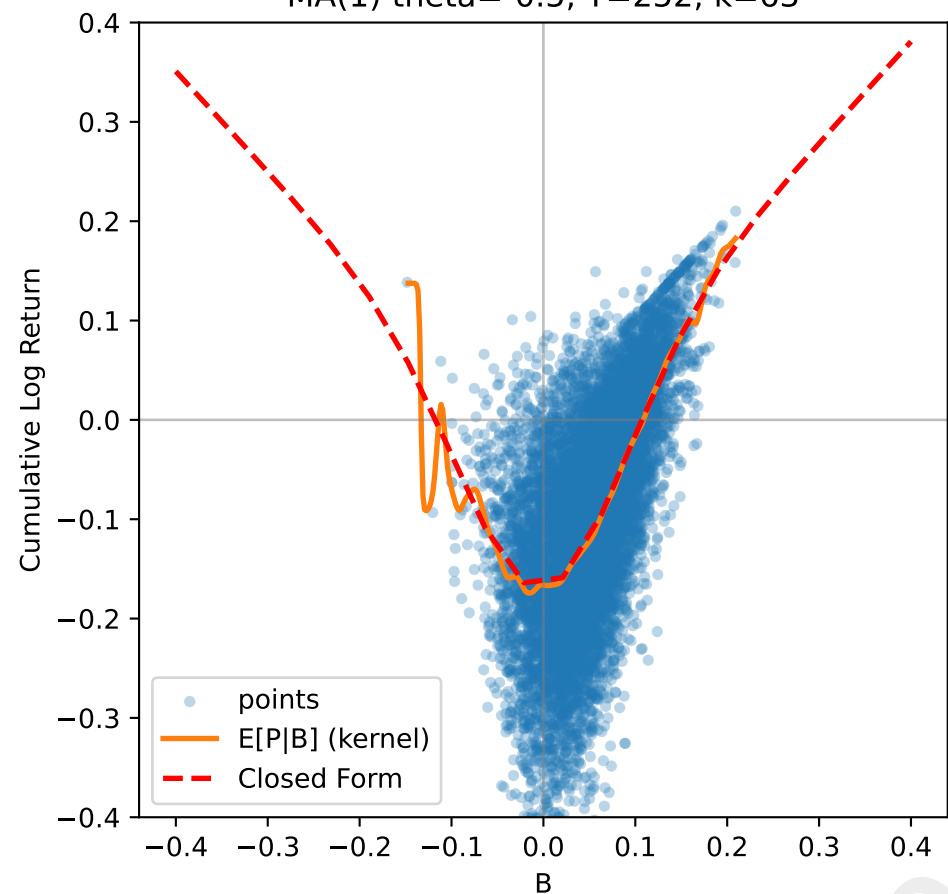
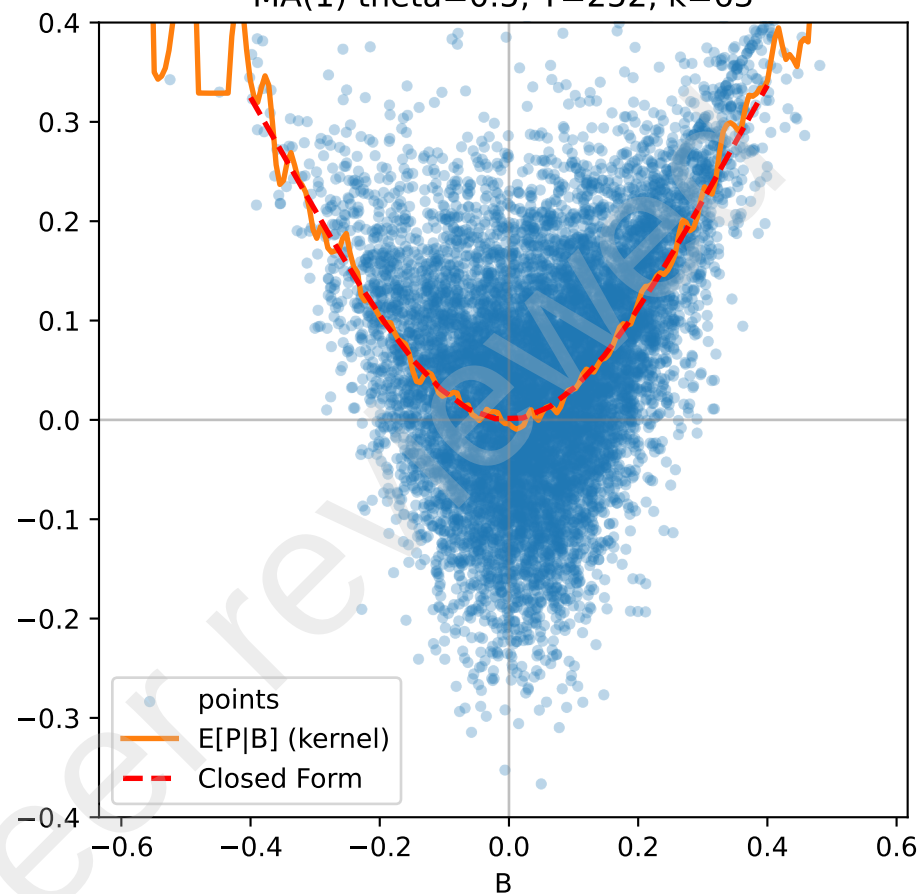
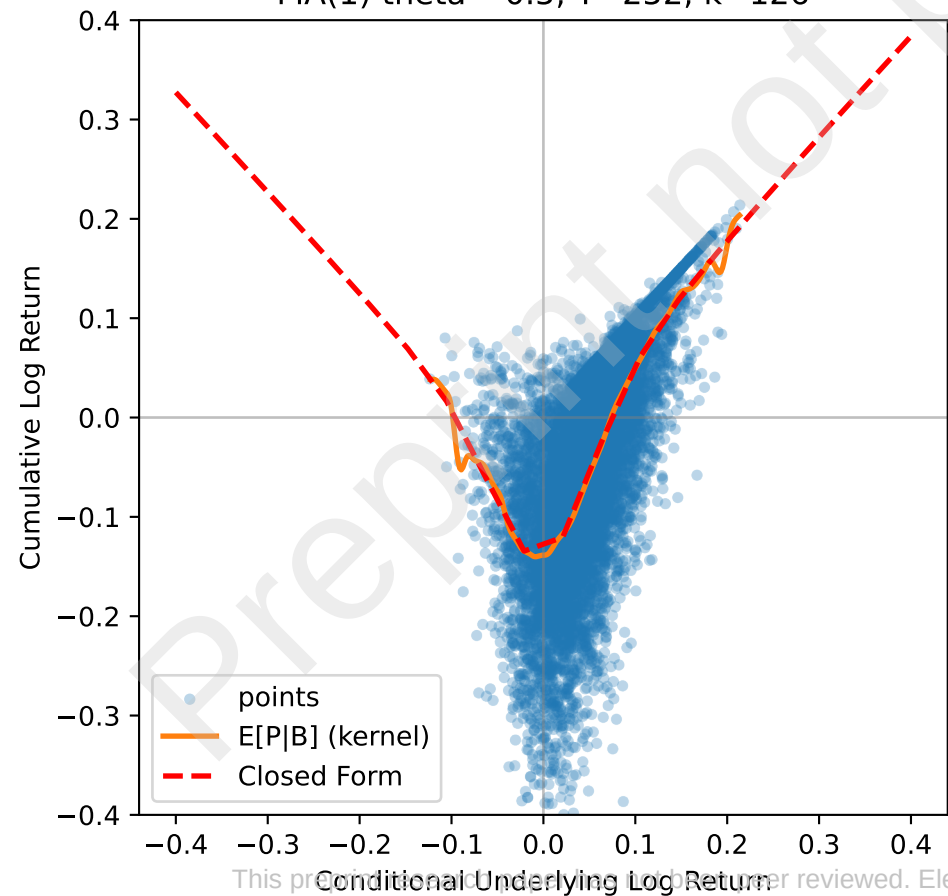
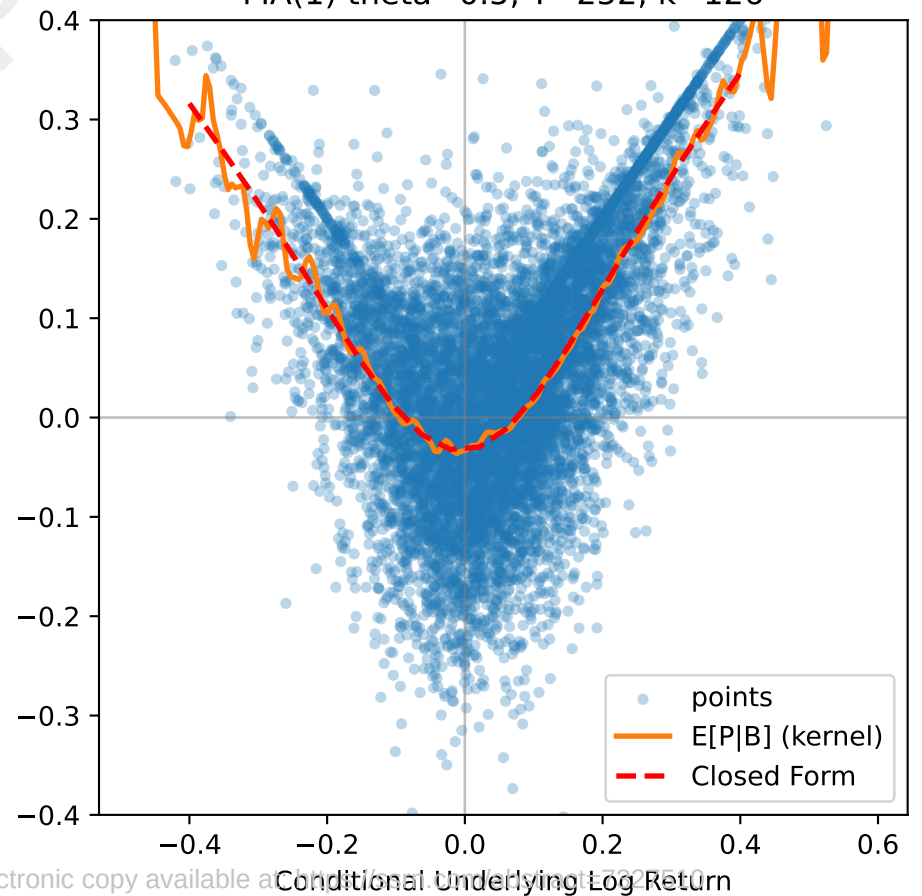
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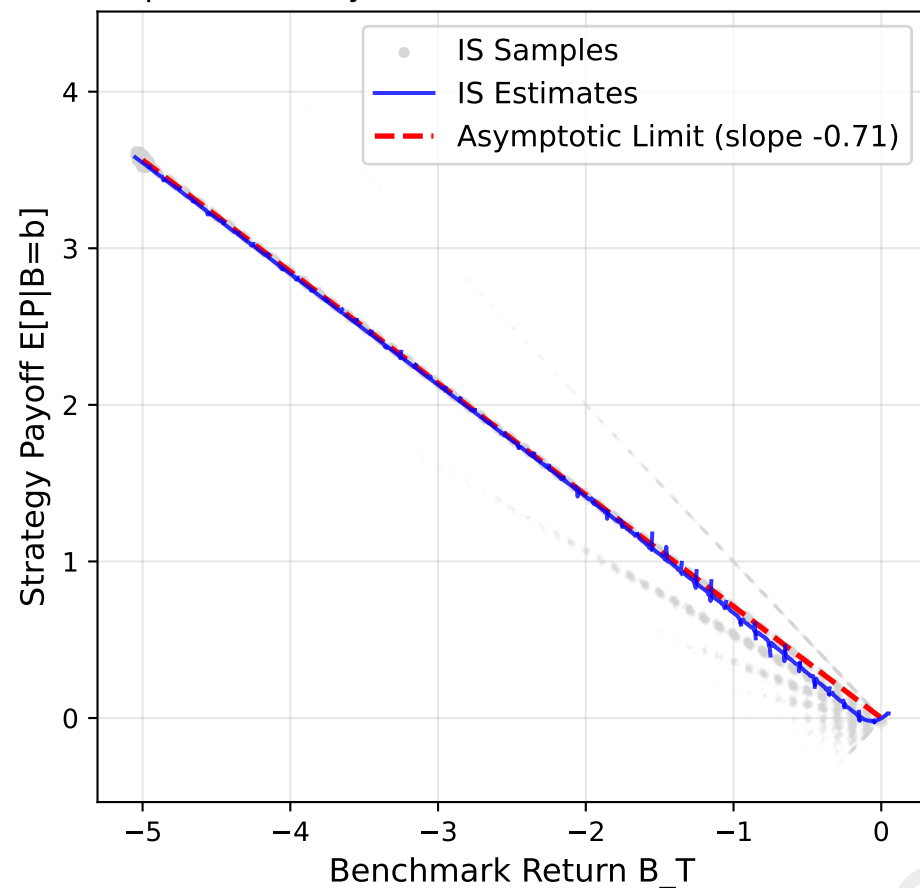
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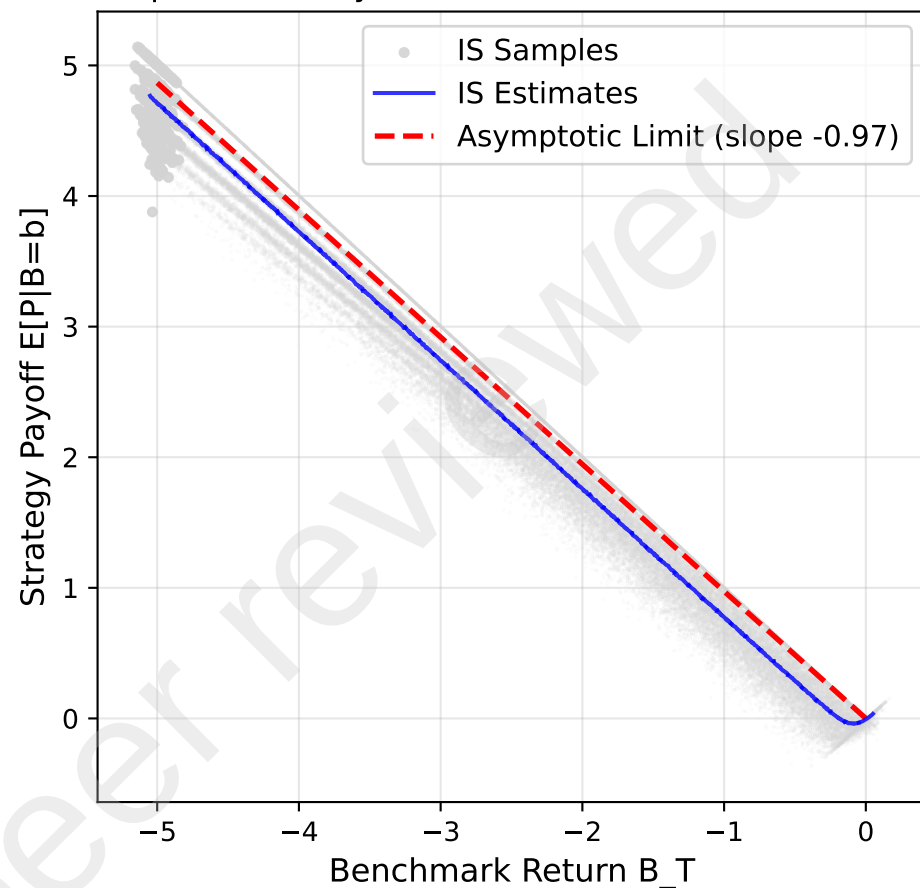


MA(1) $\theta=-0.5$, $T=252$, $k=63$ MA(1) $\theta=0.5$, $T=252$, $k=63$ MA(1) $\theta=-0.5$, $T=252$, $k=126$ MA(1) $\theta=0.5$, $T=252$, $k=126$ 

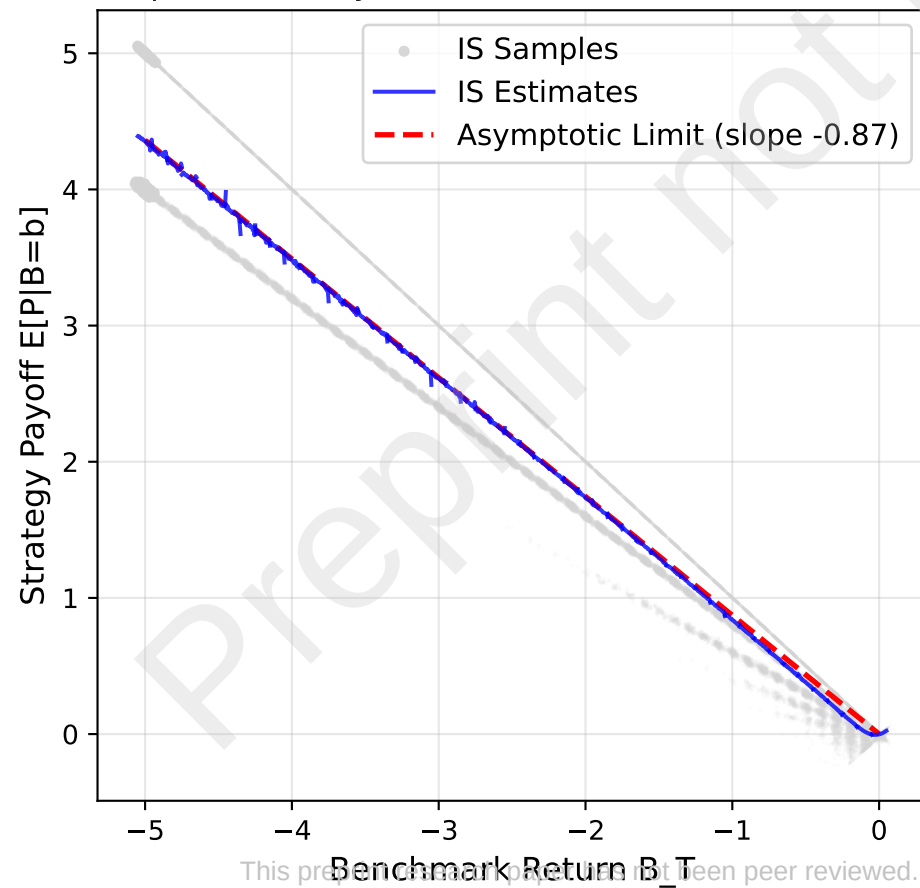
Deep Crash Analysis: MA(1) $\theta=-0.3$, $T=10$, $k=252$



Deep Crash Analysis: MA(1) $\theta=0.0$, $T=63$, $k=252$



Deep Crash Analysis: MA(1) $\theta=0.0$, $T=10$, $k=126$



Deep Crash Analysis: MA(1) $\theta=0.1$, $T=1$, $k=126$

